The Buffett Indicator: International Evidence*

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Abstract

Warren Buffett suggested that the ratio of the market value of publicly traded stocks to economic output could identify equity market mispricings. We extend the existing research from the United States to international equity markets by investigating the return-predictive characteristics of the market value of equity-to-gross domestic product for a dataset comprising fourteen countries. The findings corroborate that the "Buffet indicator" explains a large fraction of ten-year return variation for the majority of countries outside the United States. Low ratios have predicted above-average investment returns, while periods of high ratios have, on average, been followed by below-average returns over the subsequent ten-year period. We also compare the "Buffett indicator" to other well-known stock market valuation signals and phenomena, such as the cyclically-adjusted price-to-earnings ratio and mean-reversion in stock returns.

Introduction

At the turn of the millennium, Buffett wondered about the exceptional performance as measured by the Dow Jones Industrial Average during the 17-year window between 1981 and 1998 compared to the dismal performance during the equally long period between 1964 and 1981 in spite of a significantly more beneficial macroeconomic setup for the latter period, as economic output increased at more than twice the rate between 1964 and 1981 than between 1981-1998. In two Forbes interviews (Buffett and Loomis 1999, 2001), Buffett suggested that the market value of equity (MVE) scaled by gross domestic product (GDP), which subsequently will be referred to as the MVE/GDP ratio, would have heralded higher returns

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for 1981-1998 by indicating cheaper valuations for the stock market. In spite of Buffett's claim that the ratio of market capitalisation to economic output is *"probably the best single measure of where valuations stand at any given moment"*, its forecast ability has drawn comparatively little academic scrutiny.[1] Our research aims to fill this gap.

The intuition behind the MVE/GDP ratio as a valuation signal is simple. When the price of equities, as a claim on future profits, goes up without a commensurate increase in economic output, the forward earnings yield decreases, rendering, ceteris paribus, equity investments less attractive. Vis-à-vis solely price-oriented valuation techniques, such as the cyclically-adjusted price-to-earnings ratio, MVE/GDP has the advantage that its nominator contains information on quality (price) as well as quantities (equity issuance) and thereby incorporating investor sentiment not captured by purely price-based valuation metrics. New share issuance increases (decreases) when equity markets are overvalued (undervalued) if corporate managers are able to time the market; see Baker and Wurgler (2002).

The main challenge with using the MVE/GDP ratio as a signal for long-horizon market timing purposes is the lack of a theoretical fundamental level for the ratio to converge to in equilibrium. This is one of the reasons why at each point in time, there exist substantial differences for the MVE/GDP ratio across countries. For example, by year-end 2019, the ratio stood at 148% in the U.S. and only 55% in Germany. The financial structure and the regulatory environment are important factors determining the size of the stock market in a country. More specifically, the stock market of bank-based systems may reasonably be expected to be of less importance than in market-based systems, rendering a direct comparison between countries difficult. Rajan and Zingales (2003) also point out that not only is the variance of economies' equilibrium fraction of stock market capitalisation to GDP pronounced, but that there exist substantial cross-sectional differences,[2] cautioning against blindly extrapolating the findings of any country to other countries. This is one of the reasons why we look in first instance at *time-series* predictability of this valuation ratio and not at *cross-sectional* predictability from the ratio itself. However, the predicted returns from the time-series analysis will be used to determine which countries have an attractively valued stock market, and which countries have not. Hence, the trading strategy invests in countries for which the MVE/GDP ratio is low *compared to its own history*, which is not the same as an investment into a country with low MVE/GDP ratio *compared to other countries*. This direct cross-sectional comparison is not useful if fair value levels of the signal across countries are different. But even within countries, there may not necessarily exist a time-robust equilibrium value. The fair value level may change over time, as the total value of the stock market is the discounted value of future cash-flows derived from share ownership. The market value therefore increases when cash-flows are expected to have higher growth than previously predicted, or if these cash flows are discounted at a lower discount rate.

[1]The Buffet indicator is also popular among investment practitioners. For example, see the articles on the CFA blog "Enterprising Investor" by Navin Vohra (https://blogs.cfainstitute.org/investor/2021/01/29/the-buffett-indicator-revisited-market-cap-to-gdp-and-valuations), on the "Advisor Perspectives" blog by Jill Mislinski (https://www.advisorperspectives.com/dshort/updates/2021/08/05/market-cap-to-gdp-july-buffett-valuation-indicator) and on the "Fortune Financial" blog by Lawrence Hamtil (https://fortunefinancialadvisors.com/blog/applying-the-buffett-indicator-to-other-countries).

[2]See Rajan and Zingales (2003) for an explanation of factors influencing cross-country as well as time-series differences.

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The discount rate used to calculate the present value of a company's dividend stream may be affected by the long-term interest rate. Equity markets may appear expensive (cheap) when long-term interest rates are low (high). Expecting mean-reversion in the MVE/GDP ratio may therefore be partially related to an expectation of mean-reversion in long-term interest rates. Also, the willingness to be a public firm may change over time (see, Doidge, Karolyi, and Stulz (2017) and McDonald IV (2022)). Moreover, if listed companies transform to multinational companies or a stock exchange attracts foreign companies, the link with domestic economic growth weakens. Although these trends tend to be slow moving processes, they may create noise to this valuation signal. Each of these challenges make it worthwhile to examine whether the "Buffett indicator" can be used for long-horizon market timing.

The contribution of this paper is threefold. First, we analyse the international evidence of the forecasting ability of the MVE/GDP ratio for equity market returns. Our research complements existing research that is limited to the U.S. (see Lleo and Ziemba 2018 and Umlauft 2020). Our data goes back to 1973, therefore substantially extending the international study by Chang and Pak (2018) that starts in 1997.[3] Second, we compare this predictor to other well-known stock market valuation signals, such as the cyclically-adjusted price-to-earnings ratio (CAPE) of Campbell and Shiller (1988, 1998, 2001) and mean-reversion in stock returns as documented by, e.g., Poterba and Summers (1987), Fama and French (1988), Spierdijk, Bikker and Van den Hoek (2012), Moskowitz, Ooi and Pedersen (2012) and Zaremba, Kizys, and Raza (2020). Third, we provide empirical evidence for real-time profitability over the backtest period 1985-2019, supplementing the existing literature on long-run stock return predictability (see, e.g., Goyal and Welch (2008) and Diris, Palm, and Schotman (2015)).

Our findings can be summarised as follows: First, using data from the stock markets of fourteen countries with data going back to 1973, we show that the "Buffett indicator" possesses exceptional forecast qualities over longer horizons. More specifically, we find that MVE/GDP explains 83% of overall return variation, ranging from as low as 42% for Austria to 94% for Great Britain. Using an expanding window instead of a the full sample reduces the explanatory power to still considerable 64%. Secondly, in a predictive regression where we include MVE/GDP, naive mean-reversion in returns, and the cyclically-adjusted price earnings ratio, only MVE/GDP is statistically significant, suggesting that our findings are not subsumed by other popular valuation signals. Importantly, these research findings may be transported to practical application. Our backtest shows that a simple strategy based on investing in countries with the highest model-predicted returns would have yielded statistically significant and economically meaningful alpha over a corresponding buy-and-hold

[3] A notable exception is Kuvshinov and Zimmermann (2020) who analyse the development of the market capitalisation for a seventeen-country dataset between 1870 and 2016. Their study, however, deviates in important aspects from this study: The use of annual data constructed from a wide range of historical sources (primary and secondary) makes an important contribution to the study of historical market capitalisation, thereby substantiating and extending the findings of Rajan and Zingales (2003), who study the development of market capitalisation for the period 1913–1999. Although the paper provides a short analysis of forward returns on the market capitalisation to gross domestic product, that part is limited to showing that there exists a – statistically significant – inverse relationship between the market capitalisation and forward returns. In addition, the use of annual data in combination with a substantial number of assumptions, where data was not available, makes the study less relevant for practical purposes of forecasting equity returns. Moreover, the study provides only evidence for the aggregate country dataset, not for individual countries.

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benchmark, while at the same time reducing risk, i.e. with lower volatility, superior Sharpe ratio and lower maximum drawdowns. Our backtest over the period 1985-2019 shows that a simple strategy based on investing in countries with the highest model-predicted returns would have yielded statistically and economically significant alpha of 1% per annum over a corresponding buy-and-hold benchmark. At the same time, the strategy has lower risk, significantly increasing the Sharpe ratio from 0.49 to 0.57 and reducing the maximum drawdown by 4%-points.

From 1985 to 2019, the composite, an average of all ten strategies with start years between 1985 and 1994, returned 10.5%, on average, vis-à-vis the benchmark performance of 9.5%, outperforming the latter by an average of 100bps, annually. Additionally, the outperformance was accompanied by lower volatility (13.6% vs 14.2%), a superior Sharpe ratio (0.57 vs 0.49) and a c. 4%-point reduction in maximum drawdown.

The remainder of this article is structured as follows: Section 2 explains the data used for the analysis and outlines the methodology. Section 3 presents the empirical results and Section 4 performs a robustness test. In Section 5, the empirical results are discussed. Section 6 summarises the article and provides a conclusion.

Data and methodology

The countries covered in this article include OECD countries with quarterly data available for market capitalization and gross domestic product starting in 1973. The year 1973 is a natural cut-off date since data for the majority of countries starts in that year. In addition, the requirement of ten-year forward returns in order to evaluate the forecast quality means that we have only 38 years even with data going back to 1973. The 1973 cut-off excludes Norway, New Zealand, Luxemburg, Portugal, Greece, Finland, Sweden and Spain. Exclusion thus mainly affects smaller countries, the only exemption being Spain, whose data starts only in 1987. Our final sample consists of fourteen countries. Importantly, all G7 nations – Canada, France, Germany, Italy, Japan, United Kingdom, and the United States – make the cut. Smaller countries with sufficient data include Australia, Austria, Belgium, Denmark, Ireland, the Netherlands as well as Switzerland.

For the construction of the market value of equity-to-gross domestic product (MVE/GPD) ratio, the following data series were used: The nominator, or market value of equity, is based on Refinitiv's Thomson Reuters Datastream country index data, respectively, the total market series (TOTMK...) in local currencies. The denominator for the ratio is based on quarterly nominal gross domestic product (annualised) data sourced from the OECD Quarterly National Accounts statistics: Gross domestic product - expenditure approach – National currency, current prices, annual levels, seasonally adjusted (series code: B1_GE measure: CARSA).[4] Data is in local currencies.

[4]The exact ratio may not be available at the end of each quarter because GDP numbers are released with some delay. However, the variability of the Buffett indicator is mainly driven by the variability of stock market returns. Moreover, consensus estimates of GDP are typically available, further reducing uncertainty.

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For the construction of cyclically-adjusted price-earnings (CAPE, Shiller P/E, P/E10) ratios, we follow Shiller (1998). Since the construction of P/E10 ratios requires a ten-year period, data availability only starts in 1983. In addition to the ten-year lag, data for Denmark, France and Italy are further shortened due to later data availability (Denmark: 31 March 1988, France: 31 March 2000, Italy: 31 March 1996). For comparability purposes, we compute MVE/GDP figures starting from the same date in the context of comparative model evaluation in section 4. In other words, 1983 constitutes the default start date for both P/E10 as well as MVE/GDP in section 4.1. For Denmark, France and Italy, whose data availability for P/E10 data starts later (see above), we adjusted MVE/GDP computation periods accordingly as well. Returns are derived from Datastream Global Index (DSGI) data, using the total return index (RI) function of the total market (TOTMK...) series. The returns are denominated in local currencies and always include dividends.

Short (three-month) rates for computing excess returns are sourced from Global Financial Data (GFD). Missing data was complemented by three-month money market rates provided by the OECD for Austria from January 1991 onwards and from January 2009 onwards for Ireland. For Denmark and Switzerland, no three-month rates were obtainable for the beginning of the period. For Switzerland, the ten-year government bond rate was used between March 1973 and December 1979, instead. For Denmark, the policy rate of the central bank of Denmark was used for the period between March 1973 and December 1975.

We test return predictability for MVE/GDP in the spirit of Fama and French (1988) who document that the predictable component of return variation increases as the forecasting horizon expands. Given the customary use of ten-year timeframes for testing long-term return predictability, the remainder of the paper will focus on ten-year observation periods. As figure 1 shows, however, MVE/GDP is robust to other forecast horizons as well.[5]

Forward ten-year annualised returns are regressed on the starting value of the log of MVE/GDP,[6] taking the form of:

$$R_{t:t+10} = \alpha + \beta \times ln\Big(\frac{MVE_t}{GDP_t}\Big) + \epsilon_{t:t+10} \tag{1}$$

Over the last decades, the finance literature has produced substantial evidence for return predictability, particularly over long forecast horizons (see, e.g., Fernandez-Perez, Fuertes, Miffre 2017; Lioui and Poncet 2019). Recent research sheds doubt on the superiority of long-run horizons, however, arguing that the serial-correlation introduced by the use of long-term forecast horizons which frequently necessitate the use of overlapping time-periods may

[5] Average and median R^2 increase substantially up to the 11-year forecast horizon. Although the maximum R^2 is reached only for the 20-year forecast horizon, incremental increases beyond the 11-year foreacast horizon are immaterial. Given the long-term nature of the Buffett indicator, meaningful statistical robustness as well as explanatory power is achieved only towards the seven-year forecast horizon. However, even at shorter forecast horizons, MVE/GDP would have yielded impressive results. For further details, see tables 15-20 in the appendix for exemplary forecast horizons of five, seven, eight, twelve, thirteen and fifteen years.

[6] Throughout this article, MVE/GDP is computed as ln(MVE/GDP), which provides substantially better results. While we only report results for ln(MVE/GDP) in this article, computations based on MVE/GDP are available upon request.

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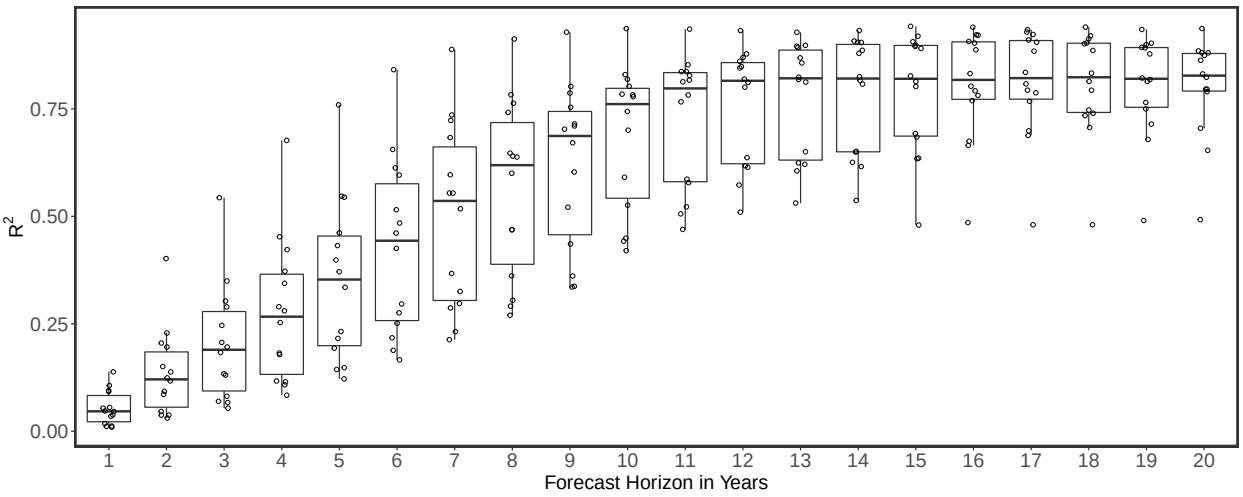


Figure 1: R^2 and forecast horizon

account for higher test-statistics by using artificially low standard errors which inflate t-statistics, leading to the rejection of H_0 in favour of H_A too often (Boudoukh, Richardson and Whitelaw (2008); Boudoukh, Israel and Whitelaw (2019)).

Indeed, the presence of overlapping time-periods introduces subtle but severe statistical problems. The common solution to this problem has traditionally been to calculate autocorrelation-robust standard errors based on Hansen and Hodrick (1980) and Newey and West (1987). As the serial correlation induced in the error terms by overlapping data is often very strong, however, the obtained robust estimators tend to have rather poor properties for finite samples. Thus, commonly used methods for autocorrelation correction may be inadequate and misleading in finite sample data.[7] Fortunately, more recent theoretical and practical analyses have come up with better procedures to correct for the bias induced by overlapping forecast horizons. In addition to the commonly used Newey and West (1987) heteroskedasticity- and autocorrelation-consistent (HAC) standard errors, we adjust for the overlap-induced bias by computing adjusted t-statistics following Hjalmarsson (2008, 2011) who demonstrates that, instead of using autocorrelation-robust standard errors, a parsimonious technique of dividing the t-statistics by the square root of the forecasting horizon yields not only approximate t-statistics but produces asymptotically exact corrections.

A similar corrective procedure has been suggested by Valkanov (2003) by scaling the OLS t-value with the square root of the sample size, hence $t/\sqrt{T}$. As Valkanov (2003) relies on the assumption that the forecasting horizon grows at the same pace as the sample size, so that the forecasting horizon remains a fixed fraction of the sample size, Hjalmarsson (2011) has more practical viability in the present context of forecasting returns in financial markets. Boudoukh, Israel, and Whitelaw (2019) also recommend this practical approach, especially because it is conservatively assuming that the explanatory variable, in our case the Buffet indicator, has a unit root, and therefore addresses the concern of autocorrelation in

[7] See, for instance, Hodrick (1992).

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the predictive variable. As an additional test-statistic, we compute t-values following Harri and Brorsen (2009) who, unlike Hjalmarsson (2011), correct for the sample size. Harri and Brorsen (2009) assess several alternative methods for correcting the overlap bias suggested since Valkanov (2003). Building upon previous literature, Harri and Brorsen (2009) propose rescaling OLS standard errors by either of two factors, depending on forecast horizon and sample size. Given the parameters of the data used in this article, we obtain rescaled standard errors for the computation of bias-corrected t-statistics by using the factor

$$SE^{HB} = SE^{OLS} \times \sqrt{\frac{2}{3} \times k \times \left(0.9 \times k \times \frac{k}{n}\right)}, \tag{2}$$

where SE^{HB} is the rescaled standard error based on Harri and Brorsen (2009), SE^{OLS} is the standard error obtained from ordinary least squares regressions, k denotes the level of aggregation (i.e. 40 for ten-year forward returns based on quarterly data) and n is the sample size.

Empirical results

We start with charting the evolution of MVE/GDP. For practical purposes, we show an equal-weighted aggregate of the fourteen-country sample (AGG) along with the United States as a yardstick. The Appendix contains figures for all individual countries. As Figure 2 makes clear, MVE/GDP is highly variable over time. It is low in the beginning of our sample, when financial development through the stock market was less common for many of the countries in our international sample, and increases dramatically towards the end of the sample period. Moreover, we see suspicious peaks around market tops, especially since the middle of the 1990s. In the remainder of the article, we evaluate whether MVE/GDP is a predictor for equity returns for our sample of international equity markets, complementing the work of Umlauft (2020) on the United States.

Figure 3 depicts graphically the relationship between MVE/GDP and subsequent returns for the equal-weighted sample and the United States, demonstrating the close inverse relationship between starting valuations and ten-year forward returns.

Another way to look at this is to examine the model-based expected stock market returns with the realised stock market returns over the subsequent ten-year period. Figure 4 displays observed trailing ten-year returns along with ten-year lagged model-predicted ten-year forward returns for the equal-weighted country aggregate and the United States. The chart illustrates that the simple model captures a significant degree of return variation, explaining approximately 83% for the period between 1973 and 2019.

As Table 1 shows, the MVE/GDP ratio has indeed a striking ability to predict future returns for the pooled-country dataset as well as for individual countries. Table 1 reports the explanatory power, R^2, of regressing ten-year forward returns on MVE/GDP by decade and

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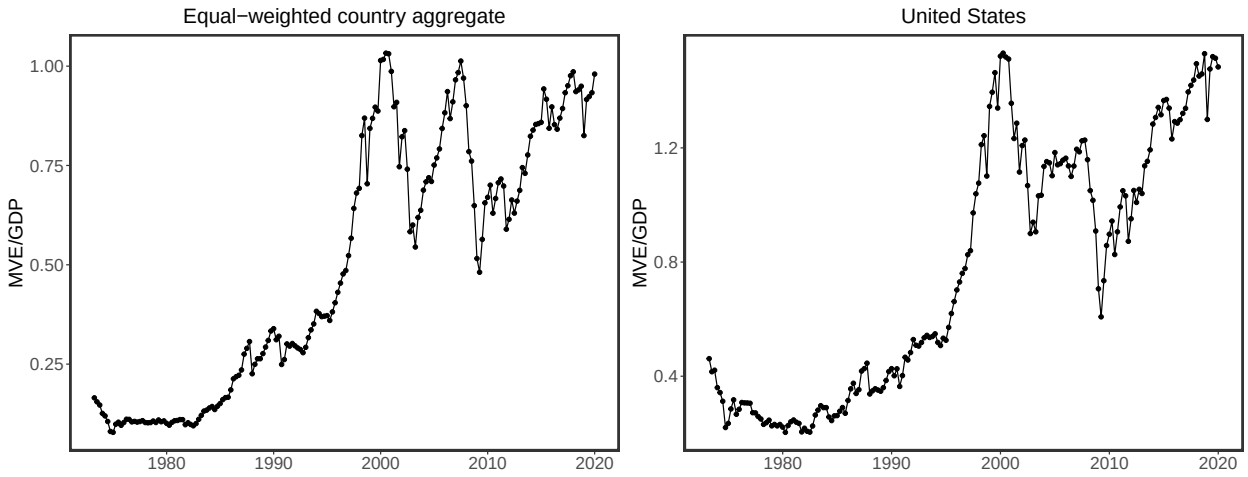


Figure 2: Market value of equity-to-gross domestic product, 1973-2019

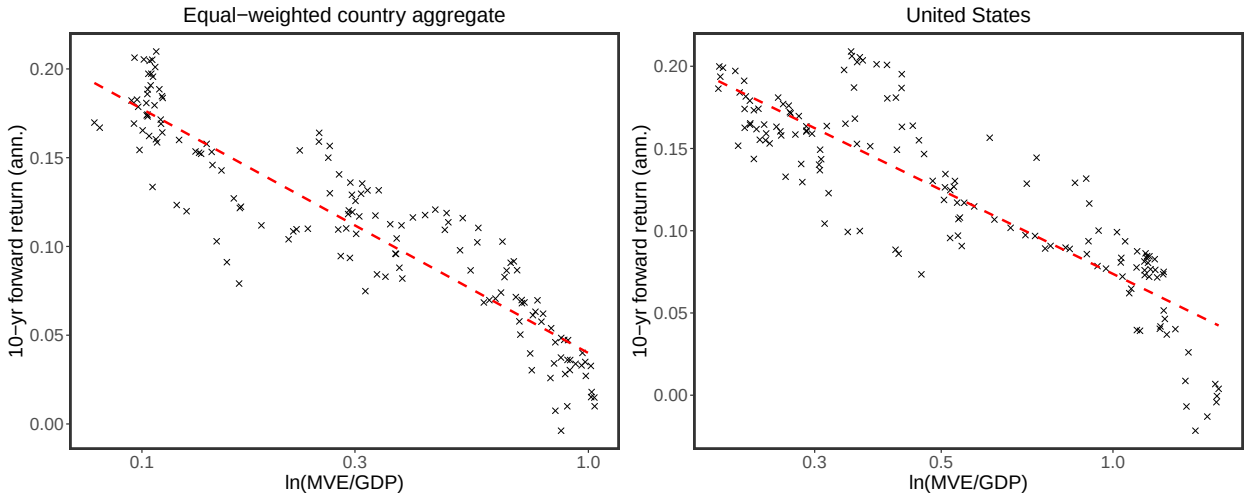


Figure 3: MVE/GDP and 10-year forward returns, 1973-2019

for the full examination period. The table makes clear that MVE/GDP explains a large component of overall return variation (83.2%) for the equal-weighted country data, although the degree itself varies over the examination period and across countries. While R^2 values can be described as robust for most countries, it exhibits considerable heterogeneity across countries, ranging from as low as 42.0% for Austria to 93.7% for Great Britain. Some points are worth mentioning in this context: First, countries with a long tradition of capital markets-based financing exhibit the highest explanatory power: Anglo-American countries, such as Great Britain, Australia, Ireland and the United States, exhibit extraordinarily high R^2 values of 93.7%, 83.0%, 78.4% and 74.4%, respectively. The only exception is Canada with a low R^2 of 44.9%. Second, with 42.0% (Austria), 52.6% (Germany) and 44.2% (Switzerland), countries which have historically been heavily dependent on bank financing show the lowest R^2.

The decade from 1980-1989 displays the lowest explanatory power, although some coun-

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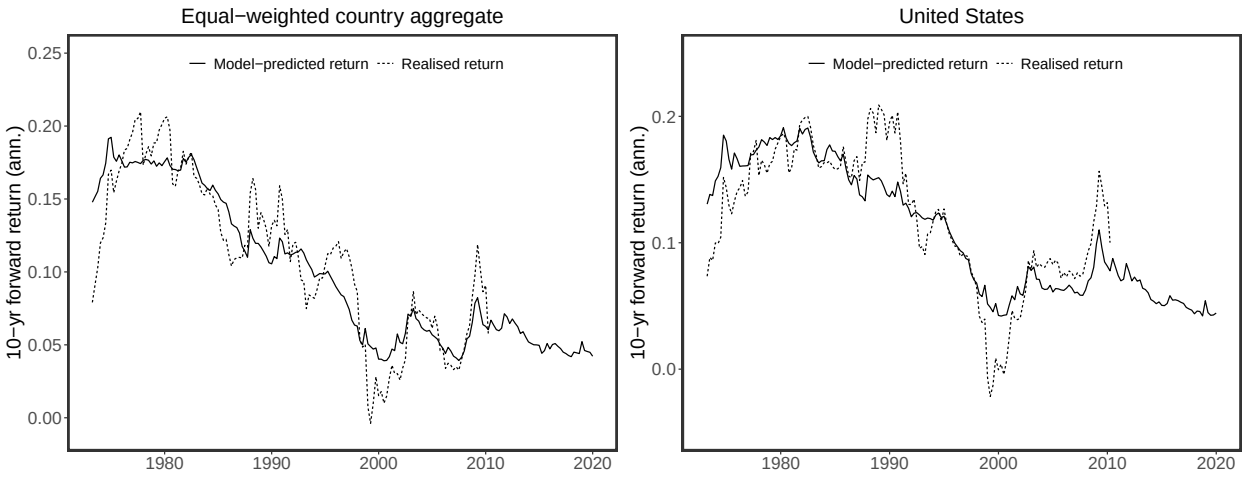


Figure 4: Realised and model-predicted 10-year forward returns

Table 1: R^2 of 10-yr forward returns regressed on ln(MVE/GDP) by decade, 1973-2019

Period	AUS	AUT	BEL	CAN	CHE	DEU	DNK	FRA	GBR	IRL	ITA	JPN	NLD	USA	AGG
1973-2009	83.0	42.0	70.1	44.9	44.2	52.6	59.1	78.3	93.7	78.4	81.9	80.2	81.9	74.4	83.2
1973-1979	65.0	70.2	85.4	3.9	15.9	15.2	60.1	62.7	83.7	47.8	82.6	26.4	82.6	79.2	46.7
1980-1989	62.6	87.3	64.4	1.2	19.1	22.6	53.3	54.8	76.0	47.1	65.8	93.5	65.8	0.6	55.5
1990-1999	45.6	17.0	59.5	63.7	82.3	53.2	42.3	67.2	78.1	80.0	65.3	13.8	65.3	88.9	63.8
2000-2009	80.6	79.0	63.0	85.2	74.4	67.5	35.5	66.9	92.5	86.7	85.0	9.8	85.0	80.4	81.5

Abbreviations: AUS = Australia, AUT = Austria, BEL = Belgium, CAN = Canada, CHE = Switzerland, DEU = Germany, DNK = Denmark, FRA = France, GBR = Great Britain, IRL = Ireland, ITA = Italy, JPN = Japan, NLD = Netherlands, USA = United States, AGG = equal-weighted country aggregate.

tries (Canada, Germany, Switzerland) exhibit low predictive power in the beginning of the examination period (1973-1979) as well. Notably, MVE/GDP for Japan has exceptionally high explanatory power for 1980-1989. This is probably due to Japan being the only country which experienced a major stock market boom in the 1980s (high MVE/GDP) which subsequently burst just at the end of the decade, leading to low returns in the ten-year period following heightened MVE/GDP ratios. On average, however, MVE/GDP captures a substantial share of return variability over the subsequent ten-year period.

Table 2 summarises mean and median as well as minimum and maximum annualised ten-year forward returns based on starting MVE/GDP valuation quintiles. As these quintiles are formed based on our full sample of MVE/GDP data, they cannot be interpreted as real time trading signals, however. Overall, these results lend credence to the high R^2 values reported above by showing that both mean and median forward returns monotonically decrease as the market value of equity relative to gross domestic product increases. Mean annual total returns for the equal-weighted country aggregate amount to 18.3% for MVE/GDP starting values in the bottom quintile, with returns lying between 12.3% and 21.0%. At the other side of the spectrum, mean returns averaged 3.6% for MVE/GDP ratios in the top quintile, with a minimum annualised return of -0.4% and a maximum annualised return of 8.7% over

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the subsequent ten-year window. Generally speaking, maximum returns exhibit quintile-based linearity comparable to mean and median returns. Minimum returns become lower as MVE/GDP increases, as well, albeit not in a comparably consistent manner as with regard to mean, median and maximum returns. All in all, these regression results provide robust evidence that MVE/GDP ratios are predictive of ten-year forward returns.

Table 2: Comparing 10-yr forward returns in % based on MVE/GDP starting quintiles

Quintile	AUS	AUT	BEL	CAN	CHE	DEU	DNK	FRA	GBR	IRL	ITA	JPN	NLD	USA	AGG
Mean returns															
1	19.3	19.9	19.4	14.0	12.2	12.6	19.4	22.1	22.9	22.6	27.2	17.6	18.8	17.1	18.3
2	14.6	7.8	11.9	10.2	12.4	10.9	14.5	15.4	17.8	19.4	13.1	7.3	15.0	16.0	13.6
3	12.4	6.9	11.6	13.2	14.3	9.2	12.7	13.3	11.9	15.9	9.6	2.1	15.4	13.5	11.5
4	10.5	9.3	10.0	10.1	8.4	8.9	11.3	9.1	8.4	8.5	6.4	0.3	7.2	9.3	8.7
5	6.6	3.2	4.3	6.3	2.8	3.7	9.1	4.0	4.3	-0.6	-0.4	-1.6	1.9	4.9	3.6
Median returns															
1	18.5	21.5	19.6	14.0	11.8	12.8	18.7	22.1	22.5	21.5	29.5	18.1	18.9	16.5	18.2
2	15.5	9.0	12.4	9.7	13.1	11.0	13.7	15.8	17.0	20.0	13.5	7.9	14.3	15.9	14.4
3	12.5	5.4	11.7	12.9	15.5	8.3	12.3	13.2	13.0	16.7	9.7	1.7	16.6	13.0	11.8
4	10.2	8.8	9.5	9.9	8.0	8.9	11.3	8.9	8.5	10.6	3.8	1.1	6.2	9.0	8.6
5	6.6	2.9	4.3	5.9	2.5	4.6	9.1	3.6	4.9	-0.5	-1.0	-1.8	2.1	7.2	3.6
Minimum returns															
1	11.4	10.9	10.6	9.7	4.8	7.2	13.6	14.3	18.6	18.3	12.5	10.0	16.3	10.4	12.3
2	9.0	-3.1	4.1	8.5	1.1	4.9	10.0	8.0	13.2	9.8	7.3	-1.1	9.2	8.6	7.9
3	9.2	-0.8	6.4	9.1	8.3	4.3	9.5	7.4	6.1	6.0	0.4	-3.0	5.5	7.3	7.5
4	6.3	3.1	3.8	6.4	2.1	2.2	3.8	2.5	6.0	-1.3	-0.5	-5.1	0.6	3.7	3.0
5	3.3	-3.0	-3.3	4.0	-0.7	-1.5	4.7	-1.1	-0.5	-6.7	-3.3	-5.9	-4.4	-2.2	-0.4
Maximum returns															
1	27.4	26.0	22.5	18.6	16.5	16.9	27.5	29.5	31.8	28.5	36.2	22.0	20.9	20.0	21.0
2	18.7	21.3	16.8	13.7	22.7	17.1	20.8	22.2	26.1	27.1	20.4	15.5	21.4	20.9	19.7
3	15.5	15.4	15.4	18.3	21.2	15.9	18.4	20.1	18.2	22.5	15.8	9.8	22.1	20.5	16.4
4	13.9	15.4	13.9	14.5	10.7	13.5	17.5	14.3	10.9	15.3	14.7	5.1	11.0	15.7	12.1
5	10.0	13.2	8.7	11.2	7.7	10.2	14.9	11.8	8.0	12.5	8.8	5.8	9.9	8.7	8.7

Abbreviations: AUS = Australia, AUT = Austria, BEL = Belgium, CAN = Canada, CHE = Switzerland, DEU = Germany, DNK = Denmark, FRA = France, GBR = Great Britain, IRL = Ireland, ITA = Italy, JPN = Japan, NLD = Netherlands, USA = United States, AGG = equal-weighted country aggregate.

After the explorative overview above, let us now turn to the evaluation of the statistical significance. Table 3 below reports R^2 values along with Newey–West's (1987) heteroskedasticity- and autocorrelation-consistent standard errors, t-statistics and p-values, as well as standard errors, t-statistics and p-values derived from bias-corrective procedures based on Harri and Brorsen (2009) and Hjalmarrson (2011). Importantly, the coefficients of all countries exhibit exceptionally high t-values, with each country showing statistical significance at the 5%-level based on the widely-used Newey and West (1987) heteroskedasticity- and autocorrelation-consistent standard errors. However, Table 3 also shows that the serial-correlation-corrective procedure of Harri and Brorsen (2009) as well as of Hjalmarrson (2011) results in markedly lower t-values than that of Newey and West (1987). Given that both Harri and Brorsen (2009) and Hjalmarrson (2011) more accurately account for finite sample properties, these results are not surprising. However, even after correcting for the overlapping time-period bias based on Harri and Brorsen (2009) and Hjalmarrson (2011), the table shows that for all

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countries, with the exception of Austria, Canada and Switzerland, the findings are statistically significant at the 5%-level. With p-values of 0.07 based on Harri and Brorsen (2009) and 0.11 based on Hjalmarrson (2011), the findings for Austria are the lowest, coinciding with the sample's lowest R^2 of 0.42. The t-statistics of -4.81 and -4.25 for the equally-weighted country aggregate indicate that the relationship is not only economically large, but also statistically significant.

Table 3: Regressing 10-yr forward nominal returns on ln(MVE/GDP), 1973-2019

	R^2	coef.	s.e.	t-stat.	p-value	s.e.$^{H+B}$	t-stat.$^{H+B}$	p-value^{H+B}	t-stat.^{H}	p-value^{H}
AUS	0.83	-0.0456	0.0037	-12.21***	0.00	0.0096	-4.77***	0.00	-4.22***	0.00
AUT	0.42	-0.0326	0.0085	-3.85***	0.00	0.0177	-1.84*	0.07	-1.63	0.11
BEL	0.70	-0.0495	0.0064	-7.74***	0.00	0.0150	-3.31***	0.00	-2.92***	0.00
CAN	0.45	-0.0252	0.0054	-4.63***	0.00	0.0129	-1.95*	0.05	-1.73*	0.09
CHE	0.44	-0.0300	0.0066	-4.51***	0.00	0.0156	-1.92*	0.06	-1.70*	0.09
DEU	0.52	-0.0395	0.0074	-5.31***	0.00	0.0174	-2.28**	0.02	-2.01**	0.05
DNK	0.59	-0.0329	0.0056	-5.88***	0.00	0.0127	-2.60**	0.01	-2.30**	0.02
FRA	0.78	-0.0471	0.0046	-10.29***	0.00	0.0115	-4.10***	0.00	-3.63***	0.00
GBR	0.94	-0.0926	0.0041	-22.77***	0.00	0.0111	-8.33***	0.00	-7.36***	0.00
IRL	0.78	-0.0889	0.0092	-9.69***	0.00	0.0216	-4.12***	0.00	-3.64***	0.00
ITA	0.82	-0.0766	0.0086	-8.94***	0.00	0.0166	-4.60***	0.00	-4.07***	0.00
JPN	0.80	-0.1327	0.0143	-9.3***	0.00	0.0305	-4.36***	0.00	-3.85***	0.00
NLD	0.78	-0.0762	0.0065	-11.82***	0.00	0.0188	-4.05***	0.00	-3.58***	0.00
USA	0.74	-0.0738	0.0092	-8.02***	0.00	0.0200	-3.68***	0.00	-3.26***	0.00
AGG	0.83	-0.0591	0.0050	-11.89***	0.00	0.0123	-4.81***	0.00	-4.25***	0.00

Notes: The t-statistics and implied p-values are calculated based on Newey–West's (1987) heteroskedasticity- and autocorrelation-consistent (HAC) standard errors. The number of lags is computed based on the non-parametric bandwidth selection procedure suggested by Newey and West (1994). Weights are computed using the Bartlett kernel as described in Newey and West (1987, 1994). Standard errors as well as t-statistics and p-values computed based on Harri and Brorsen (2009) are denoted by superscript $^{H+B}$, and t-statistics and p-values based on Hjalmarsson (2008, 2011) are denoted by superscript ^{H}. Statistical significance at the 1% level is indicated by ***, at the 5% level by ** and at the 10% level by *. *Abbreviations:* AUS = Australia, AUT = Austria, BEL = Belgium, CAN = Canada, CHE = Switzerland, DEU = Germany, DNK = Denmark, FRA = France, GBR = Great Britain, IRL = Ireland, ITA = Italy, JPN = Japan, NLD = Netherlands, USA = United States, AGG = equal-weighted country aggregate.

As is customary in the financial economics literature, where data particularly for longer-term frequencies is scarce, the findings advanced in this paper are conducted in-sample. Goyal and Welch (2008) systematically investigate in-sample and out-of-sample performance of linear regressions that predict forward returns from academic research. The authors find that out-of-sample, most models lost statistical significance vis-à-vis in-sample modelling. The authors maintain, however, that some models with very-long term forecast horizons deserve more investigation – a criterium which the MVE/GDP ratio clearly meets. Table 4 shows R^2 values for in-sample modelling vis-à-vis an expanding window approach which only uses data known to investors at each respective point in time. While explanatory power decreases for the expanding window regression, average R^2 decreases by only 19 percentage points, bringing R^2 to a still stellar reading of 64.2%, implying that approximately two thirds of the return variation is captured by the model.

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Table 4: Explanatory power (R^2) of full-sample versus expanding window regression

Measure	AUS	AUT	BEL	CAN	CHE	DEU	DNK	FRA	GBR	IRL	ITA	JPN	NLD	USA	AGG
Expanding window	73.2	48.5	70.6	21.6	28.3	36.2	60.2	66.9	85.9	58.2	77.8	76.0	64.1	62.2	64.2
In-sample	83.0	42.0	70.1	44.9	44.2	52.6	59.1	78.3	93.7	78.4	81.9	80.2	77.8	74.4	83.2
Difference	-9.8	6.5	0.5	-23.4	-16.0	-16.4	1.1	-11.4	-7.8	-20.2	-4.1	-4.2	-13.8	-12.2	-19.0

Abbreviations: AUS = Australia, AUT = Austria, BEL = Belgium, CAN = Canada, CHE = Switzerland, DEU = Germany, DNK = Denmark, FRA = France, GBR = Great Britain, IRL = Ireland, ITA = Italy, JPN = Japan, NLD = Netherlands, USA = United States, AGG = equal-weighted country aggregate.

Relation to other popular valuation signals

By design, the MVE/GDP ratio is constructed with a fast-changing nominator and a persistent denominator. The dominance of MVE in the MVE/GDP ratio gives rise to the possibility that the forward return estimate obtained from regressing on the MVE/GDP ratio may be driven solely by the nominator. Abstracting from new share issuance and buybacks, the increases and decreases in MVE are caused by realised stock returns. Hence, the predictive ability of the "Buffett indicator" may simply be due to mean-reversion in stock returns, i.e. high past returns are followed by low future returns, and vice versa. We investigate the relation with mean-reversion in stock returns in the next subsection. In addition, one can wonder whether the use of GDP is the most logical fundamental for comparison, as it includes economic activity that is not captured by the market for listed equity. Another popular equity market valuation measure looks at the market value of equities relative to the earnings that these companies have produced over the past ten years. In the second subsection, we therefore compare the predictive ability of the "Buffett indicator" relative to the cyclically-adjusted price-earnings ratio.

Mean-reversion in equity market returns

Given the negative correlation, MVE/GDP hence may capture a mean-reversion effect, as observed by e.g., Poterba and Summers (1987), Fama and French (1988), Spierdijk, Bikker and Van den Hoek (2012) and Zaremba, Kizys, and Raza (2020). To assess whether the MVE/GDP forecast quality is driven by (naïve) mean-reversion, we regress ten-year forward returns on five- and ten-year past returns. Table 5 below depicts R^2 values for ten-year forward returns with two different lookback windows. The lookback windows of five and ten years are aligned to our forecast horizon and show that, at least a one-size-fits-all, mean-reversion approach has limited explanatory power, explaining less than 10% and 1% of return variation, on average, for the ten- and five-year lookback window, respectively.

In view of the low explanatory power of a naïve implementation of mean-reversion, we may reject the possibility that the forecast quality of the market value of equity-to-gross domestic product ratio may, to any substantial degree, be attributable to capturing an underlying mean-reversion process. This is in-line with the mean-reversion literature. Spierdijk, Bikker and Van den Hoek (2012), for instance, document mean-reversion half-lives ranging from 2.0 to 22.6 years with significant cross-sectional as well as time-series variability. We hence can-

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Table 5: R^2 of naïve mean-reversion, regressing 10-yr forward returns on previous 5- and 10-yr returns

Measure	AUS	AUT	BEL	CAN	CHE	DEU	DNK	FRA	GBR	IRL	ITA	JPN	NLD	USA	AGG
5-yr/10-yr	1.3	35.1	7.2	10.3	1.8	14.9	19.5	0.1	22.3	3.6	0.8	0.4	0.2	1.0	0.4
10-yr/10-yr	13.0	31.0	0.6	4.9	11.0	3.9	0.3	3.6	29.5	7.0	0.7	6.5	2.4	1.4	9.5

Abbreviations: AUS = Australia, AUT = Austria, BEL = Belgium, CAN = Canada, CHE = Switzerland, DEU = Germany, DNK = Denmark, FRA = France, GBR = Great Britain, IRL = Ireland, ITA = Italy, JPN = Japan, NLD = Netherlands, USA = United States, AGG = equal-weighted country aggregate.

not expect that a naïve implementation in the form of a one-size-fits-all approach, i.e. look-back window, would significantly contribute to the forecasting quality of MVE/GDP based on a fixed ten-year forward return horizon.

Cyclically-adjusted price-earnings ratio

Over the past decades, the seminal articles of Campbell and Shiller (1988, 1998, 2001) have rendered the so-called cyclically-adjusted price-earnings ratio (alternatively, CAPE, Shiller P/E, P/E10) a de-facto benchmark for longer-term return forecasting. Hence, the following section compares the forecast quality of MVE/GDP with that of the cyclically-adjusted price-earnings ratio. Since P/E10 ratios require a ten-year formation period, data availability only starts in 1983. In addition to the ten-year lag, data for Denmark, France and Italy are further shortened due to later data availability (Denmark: 31 March 1988, France: 31 March 2000, Italy: 31 March 1996). For comparability purposes, we compute MVE/GDP figures starting from the same date. In other words, 1983 constitutes the default start date for both P/E10 as well as MVE/GDP in this section. For Denmark, France and Italy, whose data availability for P/E10 data starts later, we adjusted MVE/GDP computation start dates accordingly.

Table 6 makes clear that compared to the cyclically-adjusted price-to-earnings ratio, MVE/GDP is substantially more accurate for individual countries and for the pooled-country aggregate, almost unexceptionally exhibiting materially higher t-values. Indeed, P/E10 captures more of the return variation of only four countries: Denmark, France, Japan and the United States. The difference for two countries – the United States (0.83 vs 0.80) and Denmark (0.42 vs 0.41) – is minor, while P/E10's R^2 for France (0.83) is markedly higher than that of the MVE/GDP ratio (0.66). For Japan, the MVE/GDP's explanatory power of 0.39 is also markedly lower than that the 0.47 for the cyclically-adjusted price-to-earnings ratio. Interestingly, the United States is the only country apart from France with a substantial R^2 for P/E10 (0.83), although R^2 for MVE/GDP is high as well (0.80). Notably, the R^2 of 0.77 for the equal-weighted pooled-country aggregate for MVE/GDP compares quite favourably to the 0.27 for P/E10.

Also of note is that there exists a striking difference between the statistical significance derived from the use of commonly used Newey and West (1987) heteroskedasticity- and autocorrelation-consistent (HAC) standard errors and the procedures aimed at correcting for finite sample properties (Harri and Brorsen (2009); Hjalmarsson (2011)). While the use

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Table 6: Comparing results of 10-yr forward nominal returns regressed on ln(MVE/GDP) and P/E10, 1983-2019

	R^2	coef.	s.e.	t-stat.	p-value	s.e.$^{H+B}$	t-stat.$^{H+B}$	p-value^{H+B}	t-stat.^{H}	p-value^{H}
Market value of equity-to-gross domestic product ratio										
AUS	0.82	-0.0475	0.0037	-12.79***	0.00	0.0118	-4.01***	0.00	-3.55***	0.00
AUT	0.54	-0.0395	0.0062	-6.41***	0.00	0.0195	-2.02**	0.05	-1.79*	0.08
BEL	0.62	-0.0460	0.0076	-6.09***	0.00	0.0195	-2.36**	0.02	-2.08**	0.04
CAN	0.44	-0.0363	0.0112	-3.23***	0.00	0.0221	-1.64	0.10	-1.45	0.15
CHE	0.79	-0.0511	0.0069	-7.44***	0.00	0.0141	-3.63***	0.00	-3.21***	0.00
DEU	0.47	-0.0456	0.0117	-3.89***	0.00	0.0263	-1.74*	0.09	-1.53	0.13
DNK	0.42	-0.0390	0.0080	-4.88***	0.00	0.0271	-1.44	0.15	-1.27	0.21
FRA	0.66	-0.1170	0.0234	-5.01***	0.00	0.0747	-1.57	0.13	-1.39	0.17
GBR	0.87	-0.0914	0.0070	-13.07***	0.00	0.0191	-4.8***	0.00	-4.24***	0.00
IRL	0.73	-0.1015	0.0157	-6.48***	0.00	0.0335	-3.03***	0.00	-2.68***	0.01
ITA	0.90	-0.1242	0.0114	-10.92***	0.00	0.0318	-3.91***	0.00	-3.45***	0.00
JPN	0.39	-0.0851	0.0168	-5.06***	0.00	0.0570	-1.49	0.14	-1.32	0.19
NLD	0.74	-0.0965	0.0149	-6.46***	0.00	0.0311	-3.1***	0.00	-2.74***	0.01
USA	0.80	-0.0911	0.0130	-6.99***	0.00	0.0250	-3.65***	0.00	-3.23***	0.00
AGG	0.77	-0.0613	0.0073	-8.42***	0.00	0.0180	-3.41***	0.00	-3.01***	0.00
Cyclically-adjusted price-earnings ratio										
AUS	0.50	-0.0034	0.0007	-5.00***	0.00	0.0019	-1.79*	0.08	-1.65	0.10
AUT	0.18	-0.0014	0.0005	-2.55**	0.01	0.0016	-0.84	0.40	-0.77	0.44
BEL	0.29	-0.0031	0.0010	-3.01***	0.00	0.0027	-1.15	0.25	-1.06	0.29
CAN	0.22	-0.0020	0.0007	-2.86***	0.01	0.0022	-0.95	0.34	-0.87	0.38
CHE	0.46	-0.0031	0.0006	-5.29***	0.00	0.0019	-1.64	0.10	-1.51	0.13
DEU	0.35	-0.0027	0.0007	-4.12***	0.00	0.0021	-1.31	0.19	-1.20	0.23
DNK	0.41	-0.0028	0.0006	-4.24***	0.00	0.0021	-1.31	0.19	-1.25	0.21
FRA	0.83	-0.0027	0.0004	-7.63***	0.00	0.0014	-1.94*	0.06	-2.19**	0.03
GBR	0.28	-0.0045	0.0015	-3.08***	0.00	0.0041	-1.11	0.27	-1.02	0.31
IRL	0.30	-0.0044	0.0011	-4.06***	0.00	0.0038	-1.18	0.24	-1.08	0.28
ITA	0.15	-0.0012	0.0003	-3.99***	0.00	0.0023	-0.5	0.62	-0.52	0.60
JPN	0.47	-0.0015	0.0003	-5.11***	0.00	0.0009	-1.67*	0.10	-1.54	0.13
NLD	0.39	-0.0042	0.0008	-5.11***	0.00	0.0029	-1.42	0.16	-1.31	0.19
USA	0.83	-0.0051	0.0004	-11.53***	0.00	0.0013	-3.88***	0.00	-3.57***	0.00
AGG	0.27	-0.0027	0.0008	-3.27***	0.00	0.0025	-1.08	0.28	-1.00	0.32

Notes: The t-statistics and implied p-values are calculated based on Newey–West's (1987) heteroskedasticity- and autocorrelation-consistent (HAC) standard errors. The number of lags is computed based on the non-parametric bandwidth selection procedure suggested by Newey and West (1994). Weights are computed using the Bartlett kernel as described in Newey and West (1987, 1994). Standard errors as well as t-statistics and p-values computed based on Harri and Brorsen (2009) are denoted by superscript $^{H+B}$, and t-statistics and p-values based on Hjalmarsson (2008, 2011) are denoted by superscript ^{H}. Statistical significance at the 1% level is indicated by ***, at the 5% level by ** and at the 10% level by *. *Abbreviations:* AUS = Australia, AUT = Austria, BEL = Belgium, CAN = Canada, CHE = Switzerland, DEU = Germany, DNK = Denmark, FRA = France, GBR = Great Britain, IRL = Ireland, ITA = Italy, JPN = Japan, NLD = Netherlands, USA = United States, AGG = equal-weighted country aggregate.

of Newey and West (1987) suggests no material difference between MVE/GDP and P/E10 as predictors for forward equity returns, both the Harri and Brorsen (2009) as well as the Hjalmarsson (2011) adjustment leads P/E10 coefficients to lose statistical significance in most cases, while MVE/GDP coefficients, by and large, retain their significance. It should

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be kept in mind that Denmark, France and Italy have substantially shorter sample lengths for P/E10 (see Section 2 for further details). As MVE/GDP computations are aligned to the P/E10 data availability for the sake of comparison, this leads to lower statistical significance for MVE/GDP and, as a consequence, to a failure to reject the null hypothesis of no predictability for Denmark and France for the shortened P/E10-aligned period, although statistical significance is exceptionally high for the period 1973-2019 (see Table 3).

Also noteworthy in this context is that MVE/GDP shows substantially higher explanatory power (R^2), on average, with regard to nominal forward returns (0.77) than with excess returns (0.19) and real returns (0.67) between 1983 and 2019. For the purpose of forecasting excess returns (i.e. nominal returns in excess of the risk-free rate), the cyclically-adjusted price-to-earnings ratio appears to yield better results (see detailed results for comparison with excess returns in Tables 9-11 and real returns in Table 12-14 in the Appendix). The superior results for P/E10 for the purpose of forecasting excess returns are likely driven by the construction of the respective valuation measures. While the cyclically-adjusted price-to-earnings ratio uses real earnings in its denominator, MVE/GDP is based on nominal values and assumes a convergence to the mean in nominal, not real terms. Since excess returns are closely related to real returns, it appears coherent that P/E10 has higher explanatory power for excess returns, explaining the fact that R^2 increases from 0.27 for nominal returns to 0.36 for excess returns for P/E10. While P/E10 explains ten-year excess forward returns more accurately than MVE/GDP, it should be kept in mind that the explanatory power of MVE/GDP with respect to nominal forward returns is materially higher (0.77) during 1983-2019 than the explanatory power of P/E10 with respect to forward excess returns (0.36) over the same period.

To determine which of the two measures drives predictability, we regress ten-year forward returns on the log of MVE/GDP and the cyclically-adjusted price-earnings ratio. Although past returns, as shown in Table 5, do not appear to meaningfully explain future returns in and of itself, we include trailing ten-year returns as an additional factor in the multiple regression framework to check whether, in combination with MVE/GDP and P/E10, they may contribute to forecasting future returns.

The regression takes the following form and the results are presented in Table 7.

$$R_{t:t+10} = \alpha + \beta \times ln\Big(\frac{MVE_t}{GDP_t}\Big) + \gamma \times P/E10_t + \delta \times RETURN_{t-10:t} + \epsilon_{t:t+10} \qquad (3)$$

Table 7 reports adjusted R^2 as well as coefficients, t-statistics and p-values for MVE/GDP, P/E10 and trailing ten-year returns. In most cases, the explanatory power (see column 2) increases substantially vis-à-vis the simple linear regression results presented in Table 6 above. Interestingly, the MVE/GDP coefficients retain statistical significance for all countries including the pooled-country aggregate, with the exception of Japan. It should be kept in mind, however, that these results are based on Newey and West (1993) standard errors, since Harri and Brorsen (2009) as well as Hjalmarsson (2011) may not be used for multiple linear regressions. The results serve to demonstrate, however, that MVE/GDP constitutes the dominant factor when combined with the cyclically-adjusted price-to-earnings ratio as

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Table 7: Regressing 10-yr forward nominal returns on ln(MVE/GDP), P/E10 and 10-yr trailing returns, 1983-2019

		MVE/GDP				P/E10				TRAILING 10Y RET.			
	Adj. R^2	coef.	s.e.	*t*-stat.	*p*-value	coef.	s.e.	*t*-stat.	*p*-value	coef.	s.e.	*t*-stat.	*p*-value
AUS	0.86	-0.0635	0.0085	-7.46***	0.00	0.0010	0.00	1.63	0.11	-0.2666	0.0839	-3.18***	0.00
AUT	0.74	-0.0371	0.0044	-8.45***	0.00	-0.0008	0.00	-3.19***	0.00	-0.1770	0.0860	-2.06**	0.04
BEL	0.70	-0.0366	0.0059	-6.25***	0.00	-0.0020	0.00	-2.35**	0.02	0.0838	0.0982	0.85	0.40
CAN	0.58	-0.0517	0.0130	-3.97***	0.00	0.0015	0.00	1.90*	0.06	-0.5528	0.1665	-3.32***	0.00
CHE	0.82	-0.0449	0.0086	-5.23***	0.00	-0.0007	0.00	-0.97	0.34	-0.0599	0.1707	-0.35	0.73
DEU	0.60	-0.0367	0.0095	-3.85***	0.00	-0.0016	0.00	-2.07**	0.04	-0.0358	0.1765	-0.2	0.84
DNK	0.59	-0.0306	0.0106	-2.88***	0.01	-0.0017	0.00	-2.04**	0.04	-0.0723	0.1225	-0.59	0.56
FRA	0.94	-0.0391	0.0068	-5.72***	0.00	-0.0012	0.00	-5.24***	0.00	-0.2488	0.0513	-4.85***	0.00
GBR	0.88	-0.1256	0.0263	-4.78***	0.00	0.0022	0.00	1.30	0.20	-0.2401	0.1519	-1.58	0.12
IRL	0.74	-0.1094	0.0231	-4.73***	0.00	0.0008	0.00	0.51	0.61	0.0758	0.1375	0.55	0.58
ITA	0.91	-0.1271	0.0125	-10.16***	0.00	0.0004	0.00	2.00*	0.05	-0.0711	0.0806	-0.88	0.38
JPN	0.51	-0.0127	0.0444	-0.29	0.78	-0.0015	0.00	-1.64	0.10	0.0859	0.1222	0.7	0.48
NLD	0.77	-0.0839	0.0223	-3.75***	0.00	-0.0010	0.00	-0.67	0.50	0.2564	0.1610	1.59	0.11
USA	0.90	-0.0147	0.0171	-0.86	0.39	-0.0049	0.00	-4.52***	0.00	0.2356	0.1467	1.61	0.11
AGG	0.79	-0.0673	0.0130	-5.16***	0.00	0.0002	0.00	0.13	0.89	-0.1444	0.1813	-0.8	0.43

Notes: The *t*-statistics and implied *p*-values are calculated based on Newey–West's (1987) heteroskedasticity- and autocorrelation-consistent (HAC) standard errors. The number of lags is computed based on the non-parametric bandwidth selection procedure suggested by Newey and West (1994). Weights are computed using the Bartlett kernel as described in Newey and West (1987, 1994). Statistical significance at the 1% level is indicated by ***, at the 5% level by ** and at the 10% level by *. *Abbreviations:* AUS = Australia, AUT = Austria, BEL = Belgium, CAN = Canada, CHE = Switzerland, DEU = Germany, DNK = Denmark, FRA = France, GBR = Great Britain, IRL = Ireland, ITA = Italy, JPN = Japan, NLD = Netherlands, USA = United States, AGG = equal-weighted country aggregate.

well as ten-year trailing returns in a multiple regression framework. Overall, the results presented in Table 7 suggest that the forecast quality of MVE/GDP may be improved by adding P/E10 as a further explanatory variable, and possibly also trailing ten-year returns.

The ratio of market valuation of equity to economic output has a significant advantage vis-à-vis solely price-oriented valuation techniques, such as the cyclically-adjusted price-to-earnings ratio. Its nominator contains information on quality (price) as well as quantities (equity issuance). Accounting for (net) equity issuances, for instance, incorporates investor sentiment not captured by purely price-based valuation metrics. Research shows that firms are able to exploit investor exuberance by timing the market and issuing relatively more equities at overvalued prices around market tops, and thus before periods of low equity returns (Loughran, Ritter, and Rydqvist 1994, Baker and Wurgler 2000). Also, more recently, share buybacks have become a popular means to return money to shareholders (Grullon and Michaely 2002). These share repurchases may be a signal by management that the stock price is undervalued.[8] MVE/GDP, thus, adds an additional information layer with

[8]Most studies find that share buybacks are associated with positive excess returns, i.a. Peyer and Vermaelen (2009), Dittmar and Field (2015) and Manconi, Peyer and Vermaelen (2019). However, more recent research is less supportive of the long-term value creation hypothesis, arguing that excess returns may merely compensate for higher risk (Bargeron, Bonaime and Thomas (2017) and Lin, Stephens and Wu (2014)). The increased use of buybacks as a means to distribute cash to shareholders has led to an alternative version of the P/E10, which accounts for all corporate payouts to investors. For details, see Bunn and Shiller (2014) and Jivraj and Shiller (2017). Since we do not have international data on buybacks, we cannot compare the Buffett indicator with this adjusted P/E10 measure. A further improvement to the traditional P/E10

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incremental explanatory power, which may contribute to the higher explanatory power as well as statistical significance.

Practical application to investment strategies

In the following part, we transfer the empirical findings of this paper to a more practice-oriented framework, by devising a few simple trading rules. Given the high explanatory power of MVE/GDP in explaining subsequent returns, we test whether buying the countries with the highest model-predicted returns would have outperformed a simple buy-and-hold strategy. It should be noted that, unlike Lleo and Ziemba (2018), we are not concerned with forecasting short-term price-action and drawdowns. We forecast long-term equity returns at a ten-year horizon. Since, with very few exceptions, at a ten-year horizon equity returns are strictly positive, timeseries strategies which attempt to time the market by varying equity market exposure, i.e. deviate from full-allocation, are bound to underperform a buy-and-hold benchmark unless leverage is employed. A more straight-forward way of utilising the time-series return forecasts developed and tested in this paper is, therefore, the use of timeseries forecasts for the purpose of allocating to the cross-section – in this context, countries – for which the timeseries forecasts are most positive and excluding those with the least attractive return predictions. It should also be noted that this is not a cross-sectional approach in the traditional sense. In the introduction, we caution against comparing MVE/GDP ratios across countries for a variety of reasons. We do not, however, allocate to countries with the lowest MVE/GDP, as a traditional cross-sectional approach would. Rather, we use return forecasts derived from timeseries analysis for the purpose of evaluating which countries are expected to show the highest returns over subsequent ten-year time-period. A strategy based on timeseries forecasts in the context of a cross-sectional approach holds the advantage of making use of the extraordinarily accurate return forecasts permitted by MVE/GDP and at the same time, the approach elegantly circumvents the practical problems associated with forecasting long-term returns for asset classes with strictly positive risk premia.

The backtest is based on the following rules:

1. We rank countries in ascending order on the basis of their model-predicted ten-year forward return. We then construct a portfolio which equal-weights the upper half of the fourteen-country dataset, i.e. invest in the seven countries with the highest model-predicted ten-year forward return. We ensure that portfolio construction uses only data which was known to investors at any given point in time by using an expanding window regression framework. Our dataset starts in March 1973, and since we perform regressions with a 10-year holding period, the first prediction is only available in June 1983. However, this prediction is based on a regression with only two (overlapping) observations. We allow the model an initial stabilisation period until year-end 1985, at which point 11 quarterly observations are available to make

ratio constitutes the "Excess CAPE Yield" measure which is defined as the CAPE yield above the ten-year interest rate, as advanced by Shiller, Black and Jivraj (2020).

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the country-specific return forecast. To account for the possibility that 1985 might not be a representative year for the strategy implementation, we report results for strategy implementations starting in each of the following nine years, as well as an average of the ten strategies.

2. Every ten years, the strategy ranks countries in ascending order of their model-predicted ten-year forward returns again and allocates according to point (1) above.

Figure 5 depicts the excees return of the reference strategy from December 1985 to December 2019 over the benchmark. The strategy returned an annual average of 10.3%, while a portfolio equal-weighted in the sample dataset's countries returned 9.5%, implying an average annual strategy outperformance of approximately 86bps over the 35-year period. Also worth mentioning in this context is that the strategy earned its excess return over the buy-and-hold index with superior portfolio characteristics, i.e. (marginally) lower volatility (13.9% vs 14.2%) and, as a corollary, a higher Sharpe ratio (0.56 vs 0.49) as well as a less severe maximum drawdown (-52.1% vs -56.1%). These results can be viewed as representative for all ten tested strategies (see Table 8), which outperformed the benchmark by approximately 1%, on average. Remarkably, all portfolio characteristics (i.e. returns, volatility as well as maximum drawdown) in each single year are better for the strategy than for the corresponding benchmark. In fact, only during 129 out of the total of 3,550 months, or 3.6%, of all ten backtested strategies, the strategy trailed the benchmark. Importantly, the trading strategy is robust to the most conservative assumption regarding trading costs. Given that rebalancing is conducted only once every ten years, the maximum annual turnover is 20% of the portfolio. Since an average turnover of 20% would only be necessary in the unlikely scenario that the full portfolio had to be sold and, analogously, a new portfolio had to be created in full (i.e. there is zero overlap between old and new portfolio), the assumption meets the criterium of cautious conservativism. Assuming high transaction costs of 1%, the trading strategy would have incurred costs of only 20 basis points p.a., thus reducing the average annual alpha from 1.0% to 0.8%.[9]

[9]Stoll (1993) estimates trading costs for US equities fell from 0.95% in 1980 to 0.42% and hence during the early, more cost-intensive period of our analysis. Thus, transaction costs of 1% for the entire period in all likelihood severely over-estimate trading costs even on an international (developed markets) level.

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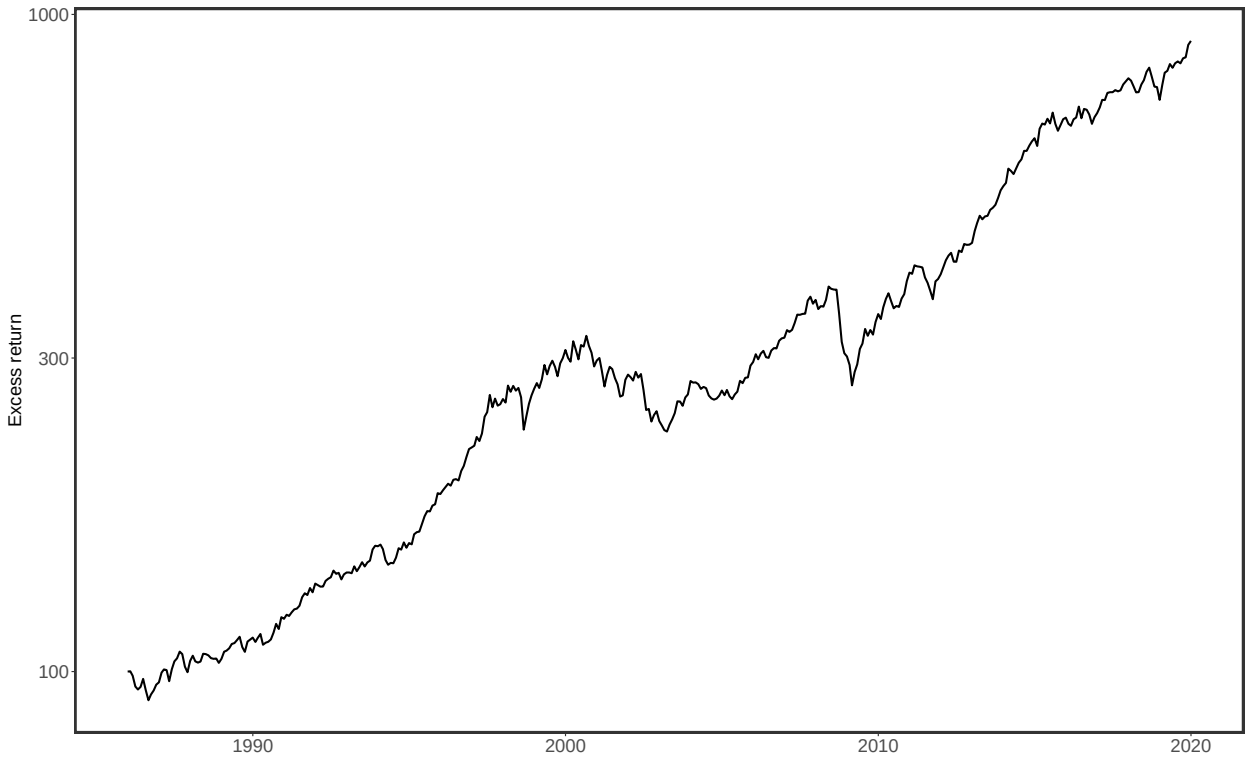


Figure 5: Strategy excess return relative to equal-weighted country index ("Composite")

Summarising, the strategy – regardless of implementation year – yielded economically meaningful alphas over a corresponding buy-and-hold benchmark equal-weighted in the dataset's fourteen countries. The composite strategy, an average of all ten strategies with start years between 1985 and 1994, returned 10.5%, on average, annually, vis-à-vis the benchmark performance of 9.5%, outperforming the latter by an average of c. 100bps per year. Additionally, the outperformance was accompanied by lower volatility (13.6% vs 14.2%), a statistically significant superior Sharpe ratio (0.57 vs 0.49) and a c. 4%-point reduction in maximum drawdown.

These findings stand in apparent contrast to Lleo and Ziemba (2018) who do not find statistical evidence for MVE/GDP as a good predictor for market crashes or corrections.[10] Given the nature of MVE/GDP, however, this is not surprising. Market crashes are, by necessity, abrupt and severe, thus the shorter-term horizon of eight quarter employed in the study. The spirit of Lleo and Ziemba (2018) is hence to be strictly distinguished from ours. We employ a ten-year forecast horizon in the spirit of Fama and French (1988) which is also frequently used in testing the forecast quality of other longer-term valuation indicators (e.g. Siegel 2016 for the cyclically-adjusted price-earnings ratio). Rather than attempting to

[10]While Lleo and Ziemba (2018) find that an unadjusted MVE/GDP ratio, as used in this article, does not provide satisfactory forecast signals for market crashes, a time-varying decision rule based on a 95% one-tail confidence interval threshold and a threshold computed using Cantelli's inequality renders the ratio a statistically significant predictor of equity market corrections.

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Table 8: Backtest results

	ANN. RETURN (%)		ANN. VOLATILITY (%)		MAX. DRAWDOWN (%)		SHARPE RATIO		JOBSON-KORKIE	
Strategy start	Strategy	B/M	Strategy	B/M	Strategy	B/M	Strategy	B/M	z-stat.	p-value
1985/12	10.3	9.5	13.9	14.2	-52.3	-56.1	0.56	0.49	1.69	0.09
1986/12	9.8	8.8	13.4	14.2	-52.1	-56.1	0.54	0.45	2.19	0.03
1987/12	10.7	9.5	12.8	13.7	-51.5	-56.1	0.62	0.52	2.79	0.01
1988/12	9.8	8.9	13.0	13.8	-51.2	-56.1	0.57	0.48	2.18	0.03
1989/12	9.0	8.2	13.1	13.9	-52.3	-56.1	0.52	0.44	1.91	0.06
1990/12	10.2	9.2	13.1	13.6	-52.1	-56.1	0.61	0.53	1.98	0.05
1991/12	9.6	9.0	13.2	13.6	-51.4	-56.1	0.58	0.52	1.60	0.11
1992/12	9.9	9.3	13.5	13.7	-52.1	-56.1	0.59	0.54	1.44	0.15
1993/12	8.6	8.3	13.3	13.8	-52.2	-56.1	0.51	0.48	1.01	0.31
1994/12	9.5	8.8	13.4	13.9	-54.3	-15.9	0.58	0.51	1.70	0.09
Composite	10.5	9.5	13.6	14.2	-52.1	-56.1	0.57	0.49	2.27	0.02

Notes: The table reports strategy results for strategies starting each year between 1985/12 and 1994/12. The average of all strategies is reported in the last row titled "Composite". The strategy's benchmark ("B/M") is an index equal-weighted at formation in the dataset's 14 countries, Australia, Austria, Belgium, Canada, Denmark, France, Germany, Great Britain, Ireland, Italy, Japan, Netherlands, Switzerland and the United States. The "Jobson-Korkie" columns report the results of the Jobson-Korkie test based on Jobson and Korkie (1981) and the Memmel (2003) correction of statistically significant difference between the Sharpe ratios of the strategy and the benchmark.

time short-term market gyrations, or indeed crashes, the ten-year horizon was chosen so as to comprise ideally two, but at least one, business cycles, and is thus concerned with forecasting long-term average returns, instead of shorter-term market movements. The findings of Lleo and Ziemba (2018) are, in fact, corroborated by Umlauft (2020) who documents high R^2 values for longer-term forecast horizons based on MVE/GDP in the United States, but low explanatory power at shorter frequencies.

MVE/GDP in historical context

Although the MVE/GDP ratio tends to exhibit a considerable degree of mean-reversion, its tendency to mean-revert has arguably declined in the recent past. In this context, it is worth noting that a regime shift appears to have manifested around the turn of the millennium, as virtually all MVE/GDP readings in the upper quintile for the country-average occurred in the period since around 2000, coinciding with the internet bubble of the late-1990s and the subprime bubble in the early-2000s as well as the subsequent recovery.

The possibility of a higher equilibrium MVE/GDP ratio is in line with Carlson, Pelz and Wohar (2002) who employ breakpoint tests on the means of valuation ratios, providing statistical evidence for an upward shift of average price-earnings ratios. Similarly, Lansing (2017) shows that lower risk-free rates, as measured by r-star, along with the growth rate of potential GDP and the core PCE inflation rate explains a significant share of higher CAPE ratios since the 1980s. Changes in equilibrium valuation levels are not confined to equity markets, however. Knoll, Schularick and Steger (2017), for instance, document a structural break in house prices induced by the rising cost of land in the second half of the twentieth century. In this context it should be noted that the time-series analysis employed in the present study omits a significant amount of twentieth century financial data for which quarterly data is not available. As Rajan and Zingales (2003) show, countries have experienced a high time-series

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variation in financial development, as proxied by the fraction of stock market capitalisation to GDP. The main finding of their "great reversals" hypothesis suggests that countries with highly developed financial sectors were more advanced as measured by market capitalisation in 1913 than they were in 1980. It was only by the end of the 1990s that the levels reached in 1913 were exceeded. Before this background we caution that the sample period studied herein might not be fully representative for the twentieth century, as a dominating period of the second half of the twentieth century was characterised by what appears to have been a catch-up to its 1913 levels, for which data is not available. The high MVE/GDP values since the 1990s might therefore not be as exceptional as they might appear at first sight. The possibility that a new equilibrium has formed towards the end of the twentieth century is substantiated by Orhangazi (2008) and Fasianos, Guevara and Pierros (2016) who demonstrate that the era of "financialisation" led to an increased importance of the financial sector and financial markets starting in the 1970s – just as MVE/GDP has begun its long, and steady march up from approximately 0.1 in the 1970s and early-1980s to almost 0.80 at the height of the internet bubble. Neither during the subsequent collapse nor at any time since did the MVE/GDP ratio approach the relatively low readings experienced in the 1970s, possibly indicating the establishment of a permanently higher equilibrium ratio.

Conclusion

This paper showed that the market value of equities scaled by gross domestic product possesses statistically significant forecast properties for long-term equity returns for our fourteen-country dataset, comparable in quality to that of the United States. With R^2 values ranging from 42.0% to 93.7%, the explanatory power of MVE/GDP for forward returns in our dataset is heterogenous, but with 83.2%, on average, comparable to that for the United States.

The relationship between the market value of equity and economic output is a crude, but straight-forward way of measuring the degree of resources directed to capital markets vis-à-vis the 'real' economy. MVE/GDP can thus be viewed as a yardstick to investor sentiment towards stock markets and, by logical extension, towards financial assets in general. As a main advantage over purely price-based valuation measures, MVE/GDP incorporates a qualitative (i.e. price) as well as a quantitative (i.e. number of shares) dimension. As an increasing amount of economic resources is being deployed on capital markets, equity prices are being driven up without a commensurate increase in "real" economic activity and expected returns fall. If share prices scaled by GDP fall, on the other hand, expected returns increase.

The analysis conducted in this paper demonstrated that the market value of equity relative to gross domestic product provides a useful tool for investors. The metric possesses strong and robust predictive properties over longer horizons (c. ten years), as market valuations relative to economic output have tended to mean-revert to a long-term equilibrium since 1973, although the latter half of the observation period has witnessed higher MVE/GDP ratios than the period until the turn of the millennium. Further research is required to determine whether this represents a temporary deviation from a long-term equilibrium, or, in fact, may mark a paradigm shift.

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Given the forecast quality of the market value of equity-to-gross domestic product ratio as demonstrated in this article, MVE/GDP constitutes an additional input factor for the purpose of equity market valuation and the formation of long-term return expectations. MVE/GDP should not replace conventional valuation measures, such as the cyclically-adjusted price-to-earnings ratio, however. As Campbell and Shiller (1998) point out, conventional valuation measures (i.e. price-to dividend and cyclically-adjusted price-to-earnings ratio) deserve a special place among forecasting variables due to the high-quality and long availability of data, going back to 1871 for the United States. Rather, MVE/GDP should be seen as a complementary valuation tool by providing incremental information on a non-earnings-based basis. The demonstrably higher explanatory power of P/E10 for the purpose of forecasting excess returns makes a convincing case for using MVE/GDP and P/E10 concomitantly.

Given current valuation levels, the likelihood of subdued returns or pronounced market losses over the next ten years is substantial. MVE/GDP thus corroborates the somewhat sobering return expectations derived from the (raw) Shiller P/E, according to which forward returns will be an order of magnitude smaller than they have been over the past decade. MVE/GDP lends further support to the depressed longer-term return forecasts derived from elevated Shiller P/E ratios. Valuation levels comparable to those today have led to annualised forward returns over the subsequent ten years in the lower single-digit range, and occasionally even in negative territory. Current valuations certainly warrant caution, although depressed levels of interest rates globally might prove continuously supportive of equity prices in spite of valuation levels in the historical upper extreme.

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APPENDIX

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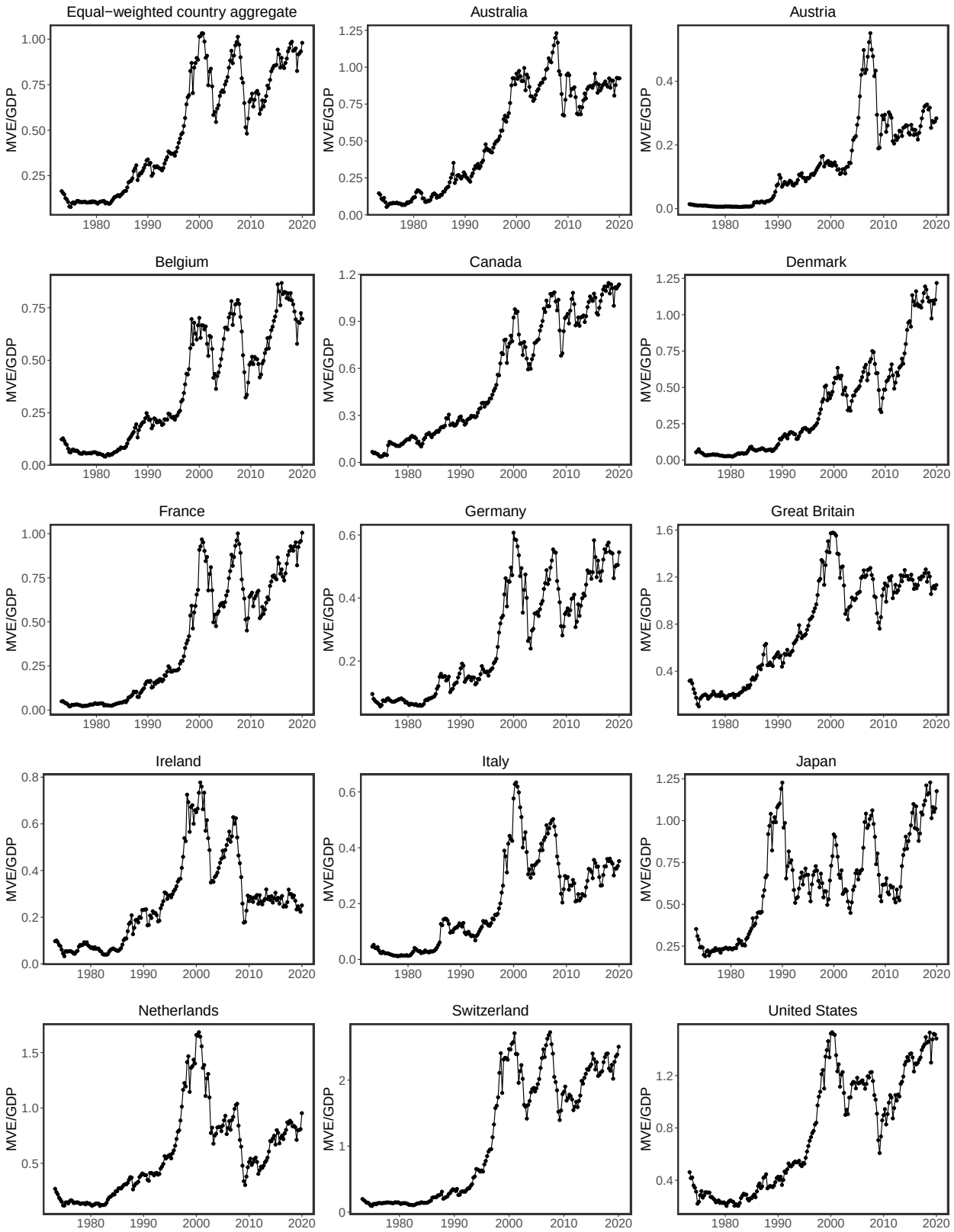


Figure 6: Market value of equity-to-gross domestic product (MVE/GDP), 1973-2019

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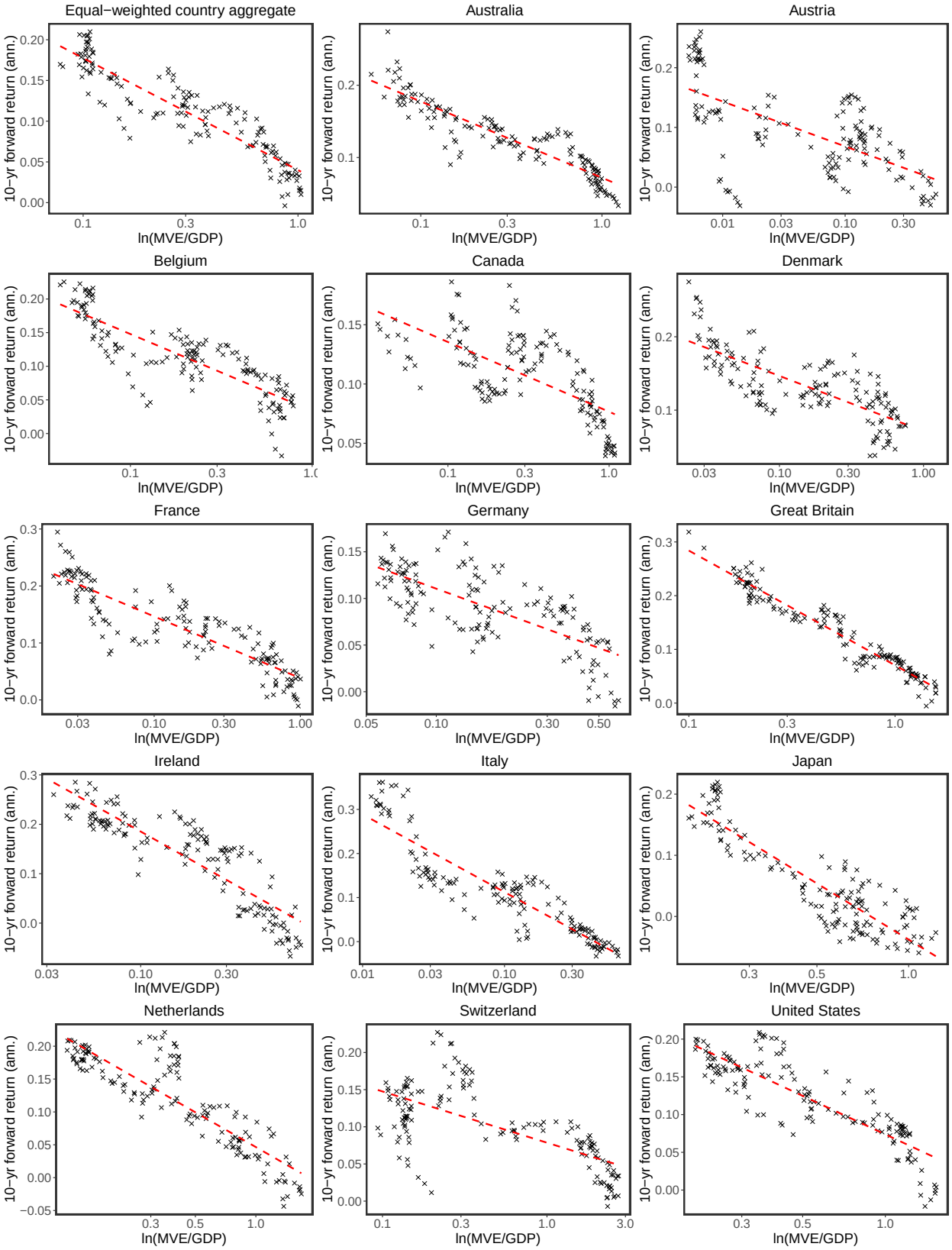


Figure 7: MVE/GDP and forward returns, 1973-2019

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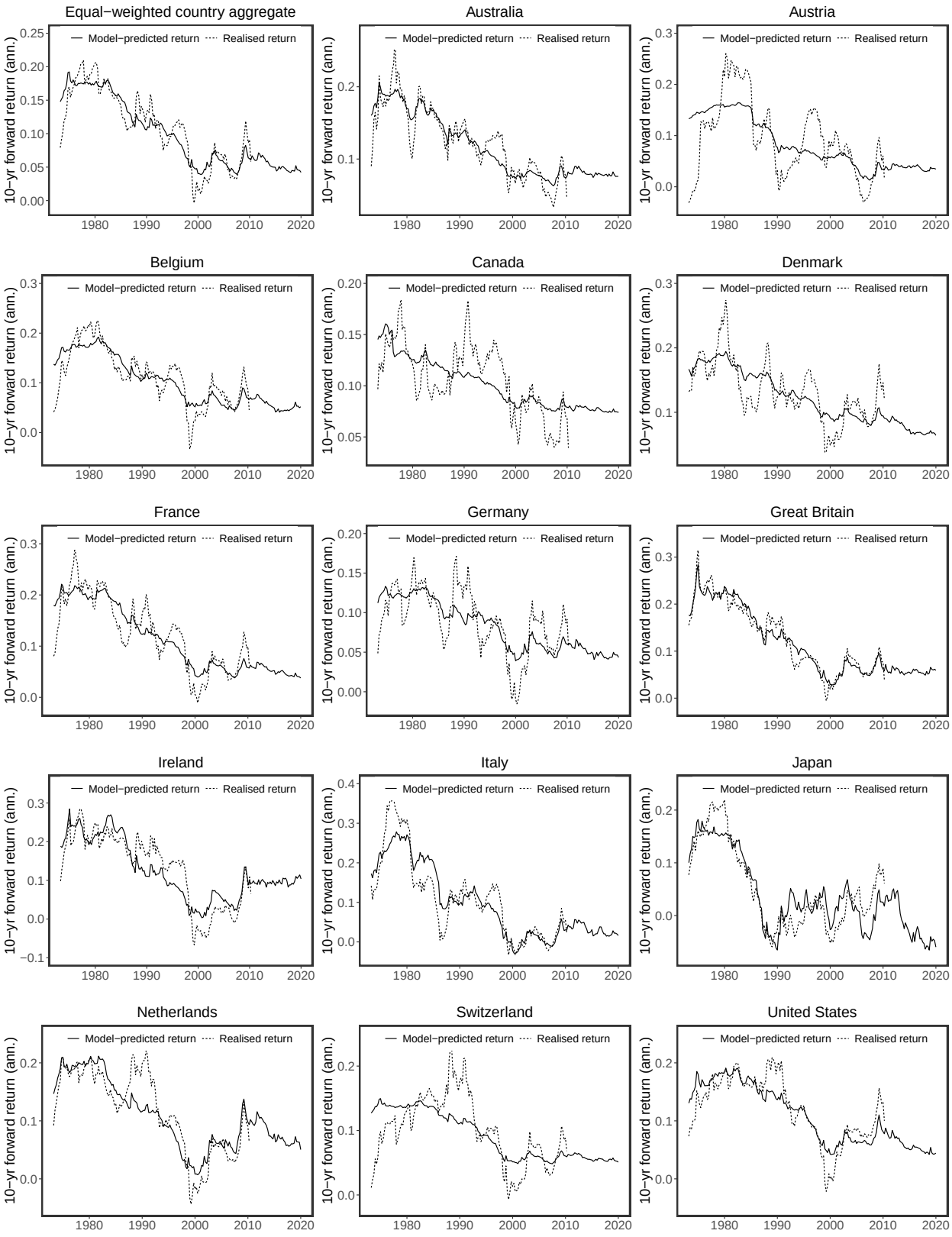


Figure 8: Realised and model-predicted 10-year forward returns

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Table 9: Regressing 10-yr forward excess returns on ln(MVE/GDP), 1973-2019

	R^2	coef.	s.e.	t-stat.	p-value	s.e.$^{H+B}$	t-stat.$^{H+B}$	p-value^{H+B}	t-stat.^{H}	p-value^{H}
AUS	0.10	-0.0102	0.0049	-2.08**	0.04	0.0134	-0.76	0.45	-0.67	0.50
AUT	0.10	-0.0143	0.0092	-1.55	0.12	0.0193	-0.74	0.46	-0.65	0.51
BEL	0.06	-0.0113	0.0082	-1.37	0.17	0.0191	-0.59	0.55	-0.52	0.60
CAN	0.18	0.0163	0.0051	3.18***	0.00	0.0159	1.03	0.31	0.91	0.37
CHE	0.14	-0.0143	0.0069	-2.08**	0.04	0.0158	-0.91	0.36	-0.80	0.42
DEU	0.08	-0.0138	0.0097	-1.42	0.16	0.0201	-0.68	0.50	-0.60	0.55
DNK	0.26	0.0226	0.0072	3.14***	0.00	0.0173	1.30	0.20	1.15	0.25
FRA	0.22	-0.0157	0.0056	-2.82***	0.01	0.0136	-1.16	0.25	-1.02	0.31
GBR	0.62	-0.0447	0.0072	-6.25***	0.00	0.0159	-2.80***	0.01	-2.48**	0.01
IRL	0.42	-0.0423	0.0098	-4.31***	0.00	0.0228	-1.85*	0.07	-1.64	0.10
ITA	0.30	-0.0308	0.0100	-3.09***	0.00	0.0215	-1.43	0.15	-1.26	0.21
JPN	0.67	-0.0999	0.0153	-6.55***	0.00	0.0321	-3.11***	0.00	-2.75***	0.01
NLD	0.60	-0.0511	0.0080	-6.36***	0.00	0.0194	-2.64***	0.01	-2.33**	0.02
USA	0.21	-0.0324	0.0138	-2.35**	0.02	0.0283	-1.14	0.26	-1.01	0.31
AGG	0.22	-0.0178	0.0065	-2.73***	0.01	0.0152	-1.17	0.24	-1.03	0.30

Notes: The t-statistics and implied p-values are calculated based on Newey–West's (1987) heteroskedasticity- and autocorrelation-consistent (HAC) standard errors. The number of lags is computed based on the non-parametric bandwidth selection procedure suggested by Newey and West (1994). Weights are computed using the Bartlett kernel as described in Newey and West (1987, 1994). Standard errors as well as t-statistics and p-values computed based on Harri and Brorsen (2009) are denoted by superscript $^{H+B}$, and t-statistics and p-values based on Hjalmarsson (2008, 2011) are denoted by superscript ^{H}. Statistical significance at the 1% level is indicated by ***, at the 5% level by ** and at the 10% level by *. *Abbreviations:* AUS = Australia, AUT = Austria, BEL = Belgium, CAN = Canada, CHE = Switzerland, DEU = Germany, DNK = Denmark, FRA = France, GBR = Great Britain, IRL = Ireland, ITA = Italy, JPN = Japan, NLD = Netherlands, USA = United States, AGG = equal-weighted country aggregate.

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Table 10: Comparing results of 10-yr forward excess returns regressed on ln(MVE/GDP) and P/E10, 1983-2019

	R^2	coef.	s.e.	t-stat.	p-value	s.e.$^{H+B}$	t-stat.$^{H+B}$	p-value^{H+B}	t-stat.^{H}	p-value^{H}
Market value of equity-to-gross domestic product ratio										
AUS	0.14	-0.0125	0.0064	-1.95*	0.05	0.0166	-0.75	0.45	-0.67	0.51
AUT	0.20	-0.0209	0.0073	-2.88***	0.00	0.0223	-0.94	0.35	-0.83	0.41
BEL	0.05	-0.0119	0.0098	-1.21	0.23	0.0252	-0.47	0.64	-0.42	0.68
CAN	0.01	0.0078	0.0142	0.55	0.58	0.0279	0.28	0.78	0.25	0.81
CHE	0.58	-0.0337	0.0076	-4.41***	0.00	0.0156	-2.17**	0.03	-1.91*	0.06
DEU	0.08	-0.0182	0.0145	-1.26	0.21	0.0311	-0.59	0.56	-0.52	0.61
DNK	0.00	-0.0065	0.0108	-0.61	0.55	0.0358	-0.18	0.86	-0.16	0.87
FRA	0.52	-0.1264	0.0357	-3.54***	0.00	0.1075	-1.18	0.25	-1.04	0.31
GBR	0.36	-0.0393	0.0115	-3.41***	0.00	0.0280	-1.41	0.16	-1.24	0.22
IRL	0.50	-0.0624	0.0168	-3.72***	0.00	0.0334	-1.87*	0.06	-1.65	0.10
ITA	0.88	-0.1170	0.0059	-19.87***	0.00	0.0319	-3.67***	0.00	-3.24***	0.00
JPN	0.25	-0.0703	0.0219	-3.21***	0.00	0.0646	-1.09	0.28	-0.96	0.34
NLD	0.61	-0.0724	0.0175	-4.13***	0.00	0.0315	-2.3**	0.02	-2.03**	0.04
USA	0.51	-0.0611	0.0161	-3.8***	0.00	0.0321	-1.91*	0.06	-1.68*	0.10
AGG	0.19	-0.0218	0.0101	-2.15**	0.03	0.0236	-0.92	0.36	-0.82	0.42
Cyclically-adjusted price-earnings ratio										
AUS	0.23	-0.0015	0.0005	-3.07***	0.00	0.0015	-0.97	0.33	-0.90	0.37
AUT	0.40	-0.0017	0.0004	-4.88***	0.00	0.0012	-1.45	0.15	-1.34	0.18
BEL	0.44	-0.0031	0.0007	-4.63***	0.00	0.0020	-1.59	0.11	-1.47	0.14
CAN	-0.01	0.0000	0.0010	-0.04	0.97	0.0023	-0.02	0.99	-0.02	0.99
CHE	0.50	-0.0025	0.0005	-5.42***	0.00	0.0014	-1.79*	0.08	-1.65	0.10
DEU	0.52	-0.0030	0.0004	-6.59***	0.00	0.0016	-1.87*	0.06	-1.72*	0.09
DNK	0.34	-0.0025	0.0008	-3.16***	0.00	0.0022	-1.13	0.26	-1.07	0.29
FRA	0.87	-0.0034	0.0004	-7.87***	0.00	0.0015	-2.29**	0.03	-2.59**	0.01
GBR	0.62	-0.0044	0.0005	-8.51***	0.00	0.0019	-2.27**	0.03	-2.09**	0.04
IRL	0.33	-0.0034	0.0008	-4.35***	0.00	0.0027	-1.27	0.21	-1.17	0.24
ITA	0.33	-0.0016	0.0003	-6.25***	0.00	0.0020	-0.8	0.43	-0.83	0.41
JPN	0.59	-0.0017	0.0002	-7.68***	0.00	0.0008	-2.14**	0.03	-1.97*	0.05
NLD	0.49	-0.0039	0.0006	-6.05***	0.00	0.0022	-1.73*	0.09	-1.60	0.11
USA	0.83	-0.0042	0.0004	-10.14***	0.00	0.0011	-3.9***	0.00	-3.59***	0.00
AGG	0.36	-0.0022	0.0006	-3.70***	0.00	0.0017	-1.33	0.19	-1.22	0.22

Notes: The t-statistics and implied p-values are calculated based on Newey–West's (1987) heteroskedasticity- and autocorrelation-consistent (HAC) standard errors. The number of lags is computed based on the non-parametric bandwidth selection procedure suggested by Newey and West (1994). Weights are computed using the Bartlett kernel as described in Newey and West (1987, 1994). Standard errors as well as t-statistics and p-values computed based on Harri and Brorsen (2009) are denoted by superscript $^{H+B}$, and t-statistics and p-values based on Hjalmarsson (2008, 2011) are denoted by superscript ^{H}. Statistical significance at the 1% level is indicated by ***, at the 5% level by ** and at the 10% level by *. *Abbreviations:* AUS = Australia, AUT = Austria, BEL = Belgium, CAN = Canada, CHE = Switzerland, DEU = Germany, DNK = Denmark, FRA = France, GBR = Great Britain, IRL = Ireland, ITA = Italy, JPN = Japan, NLD = Netherlands, USA = United States, AGG = equal-weighted country aggregate.

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Table 11: Regressing 10-yr forward excess returns on ln(MVE/GDP), P/E10 and 10-yr trailing returns, 1983-2019

		MVE/GDP				P/E10				TRAILING 10Y RET.			
	Adj. R^2	coef.	s.e.	*t*-stat.	*p*-value	coef.	s.e.	*t*-stat.	*p*-value	coef.	s.e.	*t*-stat.	*p*-value
AUS	0.31	-0.0144	0.0141	-1.02	0.31	-0.0006	0.00	-0.73	0.47	-0.2310	0.1404	-1.65	0.10
AUT	0.63	-0.0186	0.0047	-3.94***	0.00	-0.0012	0.00	-3.85***	0.00	-0.1755	0.0982	-1.79*	0.08
BEL	0.45	0.0022	0.0078	0.28	0.78	-0.0034	0.00	-3.78***	0.00	0.0455	0.1074	0.42	0.67
CAN	0.34	-0.0014	0.0162	-0.09	0.93	0.0008	0.00	0.85	0.40	-0.7462	0.1889	-3.95***	0.00
CHE	0.67	-0.0238	0.0089	-2.68***	0.01	-0.0012	0.00	-1.60	0.11	-0.0379	0.1749	-0.22	0.83
DEU	0.54	-0.0063	0.0093	-0.68	0.50	-0.0023	0.00	-2.87***	0.01	-0.1594	0.1838	-0.87	0.39
DNK	0.44	0.0008	0.0122	0.06	0.95	-0.0023	0.00	-2.26**	0.03	-0.2269	0.1433	-1.58	0.12
FRA	0.94	-0.0096	0.0082	-1.17	0.25	-0.0018	0.00	-6.52***	0.00	-0.3698	0.0620	-5.96***	0.00
GBR	0.69	-0.0520	0.0295	-1.77*	0.08	-0.0012	0.00	-0.70	0.48	-0.2417	0.1633	-1.48	0.14
IRL	0.52	-0.0600	0.0245	-2.45**	0.02	-0.0003	0.00	-0.19	0.85	-0.0655	0.1540	-0.43	0.67
ITA	0.91	-0.0885	0.0114	-7.76***	0.00	-0.0002	0.00	-0.75	0.45	-0.2169	0.0692	-3.13***	0.00
JPN	0.63	0.0308	0.0343	0.9	0.37	-0.0020	0.00	-2.48**	0.01	-0.0517	0.1109	-0.47	0.64
NLD	0.63	-0.0521	0.0258	-2.02**	0.05	-0.0015	0.00	-0.98	0.33	0.0422	0.1794	0.23	0.81
USA	0.85	0.0375	0.0192	1.96*	0.05	-0.0063	0.00	-5.43***	0.00	0.2036	0.1621	1.26	0.21
AGG	0.40	-0.0203	0.0176	-1.16	0.25	-0.0011	0.00	-0.78	0.44	-0.1616	0.2283	-0.71	0.48

Notes: The *t*-statistics and implied *p*-values are calculated based on Newey–West's (1987) heteroskedasticity- and autocorrelation-consistent (HAC) standard errors. The number of lags is computed based on the non-parametric bandwidth selection procedure suggested by Newey and West (1994). Weights are computed using the Bartlett kernel as described in Newey and West (1987, 1994). Statistical significance at the 1% level is indicated by ***, at the 5% level by ** and at the 10% level by *. *Abbreviations:* AUS = Australia, AUT = Austria, BEL = Belgium, CAN = Canada, CHE = Switzerland, DEU = Germany, DNK = Denmark, FRA = France, GBR = Great Britain, IRL = Ireland, ITA = Italy, JPN = Japan, NLD = Netherlands, USA = United States, AGG = equal-weighted country aggregate.

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Table 12: Regressing 10-yr forward real returns on ln(MVE/GDP), 1973-2019

	R^2	coef.	s.e.	t-stat.	p-value	s.e.$^{H+B}$	t-stat.$^{H+B}$	p-value^{H+B}	t-stat.^{H}	p-value^{H}
AUS	0.27	-0.0187	0.0055	-3.4***	0.00	0.0140	-1.34	0.18	-1.18	0.24
AUT	0.28	-0.0259	0.0093	-2.79***	0.01	0.0189	-1.37	0.17	-1.21	0.23
BEL	0.44	-0.0353	0.0075	-4.67***	0.00	0.0181	-1.95*	0.05	-1.72*	0.09
CAN	-0.01	-0.0012	0.0069	-0.18	0.86	0.0158	-0.08	0.94	-0.07	0.94
CHE	0.21	-0.0153	0.0061	-2.5**	0.01	0.0134	-1.14	0.25	-1.01	0.31
DEU	0.71	-0.0713	0.0084	-8.46***	0.00	0.0208	-3.43***	0.00	-3.03***	0.00
DNK	0.01	-0.0050	0.0071	-0.7	0.49	0.0164	-0.30	0.76	-0.27	0.79
FRA	0.53	-0.0431	0.0081	-5.34***	0.00	0.0188	-2.30**	0.02	-2.03**	0.04
GBR	0.47	-0.0737	0.0179	-4.13***	0.00	0.0360	-2.05**	0.04	-1.81*	0.07
IRL	0.26	-0.0297	0.0085	-3.5***	0.00	0.0226	-1.31	0.19	-1.16	0.25
ITA	0.11	-0.0148	0.0077	-1.92*	0.06	0.0188	-0.79	0.43	-0.70	0.49
JPN	0.02	0.0178	0.0228	0.78	0.44	0.0465	0.38	0.70	0.34	0.74
NLD	0.89	-0.0735	0.0030	-24.27***	0.00	0.0120	-6.11***	0.00	-5.40***	0.00
USA	0.44	-0.0520	0.0122	-4.26***	0.00	0.0270	-1.93*	0.06	-1.70*	0.09
AGG	0.49	-0.0351	0.0075	-4.7***	0.00	0.0165	-2.13**	0.03	-1.89*	0.06

Notes: The t-statistics and implied p-values are calculated based on Newey–West's (1987) heteroskedasticity- and autocorrelation-consistent (HAC) standard errors. The number of lags is computed based on the non-parametric bandwidth selection procedure suggested by Newey and West (1994). Weights are computed using the Bartlett kernel as described in Newey and West (1987, 1994). Standard errors as well as t-statistics and p-values computed based on Harri and Brorsen (2009) are denoted by superscript $^{H+B}$, and t-statistics and p-values based on Hjalmarsson (2008, 2011) are denoted by superscript ^{H}. Statistical significance at the 1% level is indicated by ***, at the 5% level by ** and at the 10% level by *. *Abbreviations:* AUS = Australia, AUT = Austria, BEL = Belgium, CAN = Canada, CHE = Switzerland, DEU = Germany, DNK = Denmark, FRA = France, GBR = Great Britain, IRL = Ireland, ITA = Italy, JPN = Japan, NLD = Netherlands, USA = United States, AGG = equal-weighted country aggregate.

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Table 13: Comparing results of 10-yr forward real returns regressed on ln(MVE/GDP) and P/E10, 1983-2019

	R^2	coef.	s.e.	t-stat.	p-value	s.e.$^{H+B}$	t-stat.$^{H+B}$	p-value^{H+B}	t-stat.^{H}	p-value^{H}
Market value of equity-to-gross domestic product ratio										
AUS	0.66	-0.0360	0.0057	-6.31***	0.00	0.0139	-2.6**	0.01	-2.30**	0.02
AUT	0.49	-0.0356	0.0062	-5.78***	0.00	0.0197	-1.81*	0.07	-1.60	0.11
BEL	0.57	-0.0426	0.0079	-5.42***	0.00	0.0198	-2.15**	0.03	-1.90*	0.06
CAN	0.27	-0.0280	0.0130	-2.16**	0.03	0.0246	-1.14	0.26	-1.01	0.32
CHE	0.05	-0.0078	0.0073	-1.06	0.29	0.0169	-0.46	0.65	-0.41	0.68
DEU	0.71	-0.0696	0.0084	-8.29***	0.00	0.0243	-2.87***	0.00	-2.53**	0.01
DNK	0.39	-0.0431	0.0115	-3.73***	0.00	0.0322	-1.34	0.18	-1.18	0.24
FRA	0.35	-0.1270	0.0494	-2.57**	0.01	0.1506	-0.84	0.40	-0.75	0.46
GBR	0.35	-0.0657	0.0177	-3.7***	0.00	0.0482	-1.36	0.18	-1.21	0.23
IRL	0.06	-0.0158	0.0101	-1.56	0.12	0.0306	-0.52	0.61	-0.46	0.65
ITA	0.57	-0.0889	0.0175	-5.08***	0.00	0.0579	-1.53	0.13	-1.36	0.18
JPN	-0.01	-0.0033	0.0458	-0.07	0.94	0.0964	-0.03	0.97	-0.03	0.98
NLD	0.82	-0.0727	0.0055	-13.25***	0.00	0.0187	-3.89***	0.00	-3.44***	0.00
USA	0.72	-0.0797	0.0142	-5.63***	0.00	0.0268	-2.98***	0.00	-2.63***	0.01
AGG	0.67	-0.0509	0.0082	-6.21***	0.00	0.0191	-2.66***	0.01	-2.35**	0.02
Cyclically-adjusted price-earnings ratio										
AUS	0.46	-0.0028	0.0005	-5.88***	0.00	0.0017	-1.66	0.10	-1.53	0.13
AUT	0.19	-0.0013	0.0005	-2.79***	0.01	0.0015	-0.88	0.38	-0.81	0.42
BEL	0.30	-0.0030	0.0010	-3.11***	0.00	0.0026	-1.17	0.24	-1.08	0.28
CAN	0.16	-0.0017	0.0008	-2.20**	0.03	0.0022	-0.78	0.44	-0.72	0.47
CHE	0.17	-0.0011	0.0005	-1.99**	0.05	0.0013	-0.83	0.41	-0.77	0.44
DEU	0.19	-0.0025	0.0010	-2.50**	0.01	0.0028	-0.88	0.38	-0.81	0.42
DNK	0.51	-0.0035	0.0008	-4.65***	0.00	0.0022	-1.6	0.11	-1.53	0.13
FRA	0.68	-0.0036	0.0009	-4.09***	0.00	0.0028	-1.29	0.20	-1.46	0.15
GBR	0.14	-0.0037	0.0014	-2.74***	0.01	0.0050	-0.74	0.46	-0.69	0.49
IRL	0.44	-0.0026	0.0006	-4.20***	0.00	0.0017	-1.57	0.12	-1.45	0.15
ITA	0.69	-0.0021	0.0003	-7.98***	0.00	0.0012	-1.7*	0.09	-1.77*	0.08
JPN	0.10	0.0009	0.0005	1.88*	0.06	0.0015	0.62	0.53	0.58	0.57
NLD	0.44	-0.0032	0.0006	-5.10***	0.00	0.0020	-1.59	0.11	-1.47	0.15
USA	0.84	-0.0047	0.0004	-11.74***	0.00	0.0011	-4.1***	0.00	-3.78***	0.00
AGG	0.29	-0.0025	0.0007	-3.60***	0.00	0.0022	-1.14	0.26	-1.05	0.30

Notes: The t-statistics and implied p-values are calculated based on Newey–West's (1987) heteroskedasticity- and autocorrelation-consistent (HAC) standard errors. The number of lags is computed based on the non-parametric bandwidth selection procedure suggested by Newey and West (1994). Weights are computed using the Bartlett kernel as described in Newey and West (1987, 1994). Standard errors as well as t-statistics and p-values computed based on Harri and Brorsen (2009) are denoted by superscript $^{H+B}$, and t-statistics and p-values based on Hjalmarsson (2008, 2011) are denoted by superscript ^{H}. Statistical significance at the 1% level is indicated by ***, at the 5% level by ** and at the 10% level by *. *Abbreviations:* AUS = Australia, AUT = Austria, BEL = Belgium, CAN = Canada, CHE = Switzerland, DEU = Germany, DNK = Denmark, FRA = France, GBR = Great Britain, IRL = Ireland, ITA = Italy, JPN = Japan, NLD = Netherlands, USA = United States, AGG = equal-weighted country aggregate.

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Table 14: Regressing 10-yr forward real returns on ln(MVE/GDP), P/E10 and 10-yr trailing returns, 1983-2019

		MVE/GDP				P/E10				TRAILING 10Y RET.			
	Adj. R^2	coef.	s.e.	*t*-stat.	*p*-value	coef.	s.e.	*t*-stat.	*p*-value	coef.	s.e.	*t*-stat.	*p*-value
AUS	0.70	-0.0434	0.0123	-3.52***	0.00	0.0002	0.00	0.30	0.76	-0.1900	0.1146	-1.66	0.10
AUT	0.70	-0.0336	0.0046	-7.32***	0.00	-0.0009	0.00	-3.36***	0.00	-0.1522	0.0877	-1.74*	0.09
BEL	0.67	-0.0321	0.0061	-5.29***	0.00	-0.0022	0.00	-2.56**	0.01	0.1043	0.0990	1.05	0.29
CAN	0.45	-0.0418	0.0147	-2.84***	0.01	0.0013	0.00	1.60	0.11	-0.6025	0.1777	-3.39***	0.00
CHE	0.18	0.0004	0.0093	0.04	0.97	-0.0010	0.00	-1.05	0.30	-0.0362	0.2087	-0.17	0.86
DEU	0.76	-0.0775	0.0060	-12.88***	0.00	0.0008	0.00	0.81	0.42	-0.4108	0.2137	-1.92*	0.06
DNK	0.76	-0.0050	0.0087	-0.58	0.56	-0.0038	0.00	-5.98***	0.00	0.3933	0.0894	4.4***	0.00
FRA	0.84	0.0215	0.0241	0.89	0.38	-0.0012	0.00	-1.43	0.16	-0.7134	0.1705	-4.18***	0.00
GBR	0.38	-0.0102	0.0369	-0.28	0.78	-0.0042	0.00	-1.46	0.15	0.3373	0.2202	1.53	0.13
IRL	0.68	-0.0003	0.0116	-0.03	0.98	-0.0019	0.00	-2.28**	0.02	-0.3242	0.0638	-5.08***	0.00
ITA	0.85	-0.0298	0.0112	-2.66**	0.01	-0.0014	0.00	-5.29***	0.00	-0.2145	0.1051	-2.04**	0.05
JPN	0.72	-0.0266	0.0339	-0.79	0.43	-0.0002	0.00	-0.23	0.82	0.5512	0.1036	5.32***	0.00
NLD	0.84	-0.0629	0.0103	-6.12***	0.00	-0.0007	0.00	-1.00	0.32	0.1794	0.0693	2.59**	0.01
USA	0.88	0.0073	0.0172	0.42	0.67	-0.0056	0.00	-5.30***	0.00	0.2602	0.1454	1.79*	0.08
AGG	0.70	-0.0535	0.0148	-3.62***	0.00	-0.0002	0.00	-0.18	0.86	-0.1168	0.1983	-0.59	0.56

Notes: The *t*-statistics and implied *p*-values are calculated based on Newey–West's (1987) heteroskedasticity- and autocorrelation-consistent (HAC) standard errors. The number of lags is computed based on the non-parametric bandwidth selection procedure suggested by Newey and West (1994). Weights are computed using the Bartlett kernel as described in Newey and West (1987, 1994). Statistical significance at the 1% level is indicated by ***, at the 5% level by ** and at the 10% level by *. *Abbreviations:* AUS = Australia, AUT = Austria, BEL = Belgium, CAN = Canada, CHE = Switzerland, DEU = Germany, DNK = Denmark, FRA = France, GBR = Great Britain, IRL = Ireland, ITA = Italy, JPN = Japan, NLD = Netherlands, USA = United States, AGG = equal-weighted country aggregate.

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Table 15: Regressing 5-yr forward nominal returns on ln(MVE/GDP), 1973-2019

	R^2	coef.	s.e.	t-stat.	p-value	s.e.$^{H+B}$	t-stat.$^{H+B}$	p-value^{H+B}	t-stat.^{H}	p-value^{H}
AUS	0.46	-0.0244	0.0048	-5.13***	0.00	0.0075	-3.24***	0.00	-2.67***	0.01
AUT	0.12	-0.0136	0.0090	-1.51	0.13	0.0105	-1.30	0.19	-1.07	0.28
BEL	0.23	-0.0241	0.0103	-2.34**	0.02	0.0125	-1.92*	0.06	-1.58	0.12
CAN	0.19	-0.0139	0.0042	-3.33***	0.00	0.0081	-1.71*	0.09	-1.41	0.16
CHE	0.14	-0.0124	0.0057	-2.18**	0.03	0.0086	-1.43	0.15	-1.18	0.24
DEU	0.21	-0.0238	0.0090	-2.64***	0.01	0.0130	-1.83*	0.07	-1.51	0.13
DNK	0.14	-0.0144	0.0070	-2.05**	0.04	0.0099	-1.46	0.15	-1.20	0.23
FRA	0.39	-0.0225	0.0050	-4.48***	0.00	0.0079	-2.84***	0.01	-2.34**	0.02
GBR	0.76	-0.0481	0.0043	-11.28***	0.00	0.0078	-6.21***	0.00	-5.12***	0.00
IRL	0.43	-0.0485	0.0114	-4.25***	0.00	0.0159	-3.05***	0.00	-2.51**	0.01
ITA	0.54	-0.0418	0.0063	-6.67***	0.00	0.0109	-3.84***	0.00	-3.16***	0.00
JPN	0.54	-0.0783	0.0160	-4.9***	0.00	0.0205	-3.82***	0.00	-3.15***	0.00
NLD	0.37	-0.0388	0.0095	-4.09***	0.00	0.0145	-2.68***	0.01	-2.21**	0.03
USA	0.33	-0.0331	0.0082	-4.02***	0.00	0.0133	-2.48**	0.01	-2.04**	0.04
AGG	0.39	-0.0285	0.0073	-3.9***	0.00	0.0101	-2.81***	0.01	-2.32**	0.02

Notes: The t-statistics and implied p-values are calculated based on Newey–West's (1987) heteroskedasticity- and autocorrelation-consistent (HAC) standard errors. The number of lags is computed based on the non-parametric bandwidth selection procedure suggested by Newey and West (1994). Weights are computed using the Bartlett kernel as described in Newey and West (1987, 1994). Standard errors as well as t-statistics and p-values computed based on Harri and Brorsen (2009) are denoted by superscript $^{H+B}$, and t-statistics and p-values based on Hjalmarsson (2008, 2011) are denoted by superscript ^{H}. Statistical significance at the 1% level is indicated by ***, at the 5% level by ** and at the 10% level by *. *Abbreviations:* AUS = Australia, AUT = Austria, BEL = Belgium, CAN = Canada, CHE = Switzerland, DEU = Germany, DNK = Denmark, FRA = France, GBR = Great Britain, IRL = Ireland, ITA = Italy, JPN = Japan, NLD = Netherlands, USA = United States, AGG = equal-weighted country aggregate.

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Table 16: Regressing 7-yr forward nominal returns on ln(MVE/GDP), 1973-2019

	R^2	coef.	s.e.	t-stat.	p-value	s.e.$^{H+B}$	t-stat.$^{H+B}$	p-value^{H+B}	t-stat.^{H}	p-value^{H}
AUS	0.72	-0.0313	0.0034	-9.3***	0.00	0.0069	-4.53***	0.00	-3.84***	0.00
AUT	0.21	-0.0213	0.0108	-1.98**	0.05	0.0146	-1.46	0.15	-1.24	0.22
BEL	0.36	-0.0329	0.0103	-3.2***	0.00	0.0154	-2.14**	0.03	-1.81*	0.07
CAN	0.28	-0.0172	0.0047	-3.7***	0.00	0.0097	-1.78*	0.08	-1.51	0.13
CHE	0.23	-0.0174	0.0060	-2.87***	0.00	0.0113	-1.54	0.13	-1.30	0.19
DEU	0.29	-0.0265	0.0074	-3.56***	0.00	0.0145	-1.82*	0.07	-1.54	0.12
DNK	0.32	-0.0205	0.0052	-3.94***	0.00	0.0105	-1.95*	0.05	-1.65	0.10
FRA	0.59	-0.0309	0.0051	-6.1***	0.00	0.0090	-3.41***	0.00	-2.89***	0.00
GBR	0.89	-0.0650	0.0029	-22.39***	0.00	0.0082	-7.92***	0.00	-6.70***	0.00
IRL	0.55	-0.0603	0.0105	-5.74***	0.00	0.0193	-3.13***	0.00	-2.65***	0.01
ITA	0.73	-0.0584	0.0072	-8.15***	0.00	0.0125	-4.68***	0.00	-3.97***	0.00
JPN	0.68	-0.1032	0.0162	-6.37***	0.00	0.0250	-4.12***	0.00	-3.49***	0.00
NLD	0.55	-0.0538	0.0072	-7.46***	0.00	0.0172	-3.13***	0.00	-2.65***	0.01
USA	0.51	-0.0474	0.0079	-5.97***	0.00	0.0163	-2.91***	0.00	-2.46**	0.01
AGG	0.60	-0.0393	0.0068	-5.79***	0.00	0.0113	-3.47***	0.00	-2.94***	0.00

Notes: The t-statistics and implied p-values are calculated based on Newey–West's (1987) heteroskedasticity- and autocorrelation-consistent (HAC) standard errors. The number of lags is computed based on the non-parametric bandwidth selection procedure suggested by Newey and West (1994). Weights are computed using the Bartlett kernel as described in Newey and West (1987, 1994). Standard errors as well as t-statistics and p-values computed based on Harri and Brorsen (2009) are denoted by superscript $^{H+B}$, and t-statistics and p-values based on Hjalmarsson (2008, 2011) are denoted by superscript ^{H}. Statistical significance at the 1% level is indicated by ***, at the 5% level by ** and at the 10% level by *. *Abbreviations:* AUS = Australia, AUT = Austria, BEL = Belgium, CAN = Canada, CHE = Switzerland, DEU = Germany, DNK = Denmark, FRA = France, GBR = Great Britain, IRL = Ireland, ITA = Italy, JPN = Japan, NLD = Netherlands, USA = United States, AGG = equal-weighted country aggregate.

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Table 17: Regressing 8-yr forward nominal returns on ln(MVE/GDP), 1973-2019

	R^2	coef.	s.e.	t-stat.	p-value	s.e.$^{H+B}$	t-stat.$^{H+B}$	p-value^{H+B}	t-stat.^{H}	p-value^{H}
AUS	0.76	-0.0347	0.0036	-9.57***	0.00	0.0076	-4.58***	0.00	-3.94***	0.00
AUT	0.27	-0.0251	0.0107	-2.34**	0.02	0.0161	-1.55	0.12	-1.33	0.18
BEL	0.47	-0.0382	0.0089	-4.3***	0.00	0.0159	-2.40**	0.02	-2.06**	0.04
CAN	0.30	-0.0193	0.0057	-3.38***	0.00	0.0114	-1.69*	0.09	-1.45	0.15
CHE	0.29	-0.0209	0.0063	-3.33***	0.00	0.0128	-1.64	0.10	-1.41	0.16
DEU	0.36	-0.0299	0.0073	-4.11***	0.00	0.0155	-1.92*	0.06	-1.65	0.10
DNK	0.47	-0.0245	0.0046	-5.3***	0.00	0.0102	-2.40**	0.02	-2.06**	0.04
FRA	0.64	-0.0358	0.0053	-6.75***	0.00	0.0103	-3.46***	0.00	-2.97***	0.00
GBR	0.91	-0.0745	0.0027	-27.54***	0.00	0.0090	-8.26***	0.00	-7.09***	0.00
IRL	0.64	-0.0674	0.0103	-6.53***	0.00	0.0199	-3.39***	0.00	-2.91***	0.00
ITA	0.78	-0.0651	0.0081	-8.02***	0.00	0.0134	-4.85***	0.00	-4.17***	0.00
JPN	0.74	-0.1144	0.0154	-7.43***	0.00	0.0264	-4.33***	0.00	-3.72***	0.00
NLD	0.64	-0.0616	0.0065	-9.52***	0.00	0.0181	-3.41***	0.00	-2.93***	0.00
USA	0.60	-0.0561	0.0078	-7.17***	0.00	0.0179	-3.13***	0.00	-2.69***	0.01
AGG	0.69	-0.0456	0.0065	-7.03***	0.00	0.0119	-3.83***	0.00	-3.29***	0.00

Notes: The t-statistics and implied p-values are calculated based on Newey–West's (1987) heteroskedasticity- and autocorrelation-consistent (HAC) standard errors. The number of lags is computed based on the non-parametric bandwidth selection procedure suggested by Newey and West (1994). Weights are computed using the Bartlett kernel as described in Newey and West (1987, 1994). Standard errors as well as t-statistics and p-values computed based on Harri and Brorsen (2009) are denoted by superscript $^{H+B}$, and t-statistics and p-values based on Hjalmarsson (2008, 2011) are denoted by superscript ^{H}. Statistical significance at the 1% level is indicated by ***, at the 5% level by ** and at the 10% level by *. *Abbreviations:* AUS = Australia, AUT = Austria, BEL = Belgium, CAN = Canada, CHE = Switzerland, DEU = Germany, DNK = Denmark, FRA = France, GBR = Great Britain, IRL = Ireland, ITA = Italy, JPN = Japan, NLD = Netherlands, USA = United States, AGG = equal-weighted country aggregate.

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Table 18: Regressing 12-yr forward nominal returns on ln(MVE/GDP), 1973-2019

	R^2	coef.	s.e.	t-stat.	p-value	s.e.$^{H+B}$	t-stat.$^{H+B}$	p-value^{H+B}	t-stat.^{H}	p-value^{H}
AUS	0.88	-0.0551	0.0031	-17.82***	0	0.0110	-4.99***	0.00	-4.55***	0.00
AUT	0.63	-0.0394	0.0057	-6.86***	0	0.0160	-2.46**	0.02	-2.24**	0.03
BEL	0.80	-0.0600	0.0053	-11.39***	0	0.0161	-3.73***	0.00	-3.40***	0.00
CAN	0.51	-0.0303	0.0052	-5.79***	0	0.0160	-1.90*	0.06	-1.73*	0.09
CHE	0.57	-0.0389	0.0070	-5.59***	0	0.0181	-2.15**	0.03	-1.96*	0.05
DEU	0.61	-0.0491	0.0073	-6.74***	0	0.0208	-2.37**	0.02	-2.16**	0.03
DNK	0.61	-0.0394	0.0061	-6.41***	0	0.0168	-2.35**	0.02	-2.14**	0.03
FRA	0.86	-0.0572	0.0043	-13.34***	0	0.0124	-4.63***	0.00	-4.22***	0.00
GBR	0.93	-0.1076	0.0061	-17.78***	0	0.0156	-6.91***	0.00	-6.30***	0.00
IRL	0.87	-0.1112	0.0082	-13.48***	0	0.0231	-4.81***	0.00	-4.38***	0.00
ITA	0.84	-0.0828	0.0086	-9.58***	0	0.0190	-4.35***	0.00	-3.97***	0.00
JPN	0.82	-0.1496	0.0176	-8.5***	0	0.0378	-3.96***	0.00	-3.61***	0.00
NLD	0.85	-0.0887	0.0069	-12.82***	0	0.0201	-4.41***	0.00	-4.02***	0.00
USA	0.81	-0.0875	0.0090	-9.69***	0	0.0227	-3.86***	0.00	-3.52***	0.00
AGG	0.89	-0.0716	0.0040	-17.97***	0	0.0136	-5.27***	0.00	-4.80***	0.00

Notes: The t-statistics and implied p-values are calculated based on Newey–West's (1987) heteroskedasticity- and autocorrelation-consistent (HAC) standard errors. The number of lags is computed based on the non-parametric bandwidth selection procedure suggested by Newey and West (1994). Weights are computed using the Bartlett kernel as described in Newey and West (1987, 1994). Standard errors as well as t-statistics and p-values computed based on Harri and Brorsen (2009) are denoted by superscript $^{H+B}$, and t-statistics and p-values based on Hjalmarsson (2008, 2011) are denoted by superscript ^{H}. Statistical significance at the 1% level is indicated by ***, at the 5% level by ** and at the 10% level by *. *Abbreviations:* AUS = Australia, AUT = Austria, BEL = Belgium, CAN = Canada, CHE = Switzerland, DEU = Germany, DNK = Denmark, FRA = France, GBR = Great Britain, IRL = Ireland, ITA = Italy, JPN = Japan, NLD = Netherlands, USA = United States, AGG = equal-weighted country aggregate.

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Table 19: Regressing 13-yr forward nominal returns on ln(MVE/GDP), 1973-2019

	R^2	coef.	s.e.	t-stat.	p-value	s.e.$^{H+B}$	t-stat.$^{H+B}$	p-value^{H+B}	t-stat.^{H}	p-value^{H}
AUS	0.90	-0.0582	0.0030	-19.54***	0	0.0114	-5.09***	0.00	-4.71***	0.00
AUT	0.65	-0.0397	0.0055	-7.22***	0	0.0168	-2.36**	0.02	-2.19**	0.03
BEL	0.81	-0.0629	0.0054	-11.7***	0	0.0174	-3.61***	0.00	-3.34***	0.00
CAN	0.53	-0.0302	0.0048	-6.26***	0	0.0164	-1.84*	0.07	-1.71*	0.09
CHE	0.60	-0.0421	0.0076	-5.55***	0	0.0196	-2.15**	0.03	-1.99**	0.05
DEU	0.62	-0.0501	0.0075	-6.72***	0	0.0224	-2.24**	0.03	-2.07**	0.04
DNK	0.62	-0.0401	0.0058	-6.88***	0	0.0181	-2.22**	0.03	-2.05**	0.04
FRA	0.90	-0.0604	0.0042	-14.29***	0	0.0117	-5.14***	0.00	-4.76***	0.00
GBR	0.93	-0.1145	0.0065	-17.66***	0	0.0183	-6.25***	0.00	-5.78***	0.00
IRL	0.89	-0.1201	0.0080	-15.04***	0	0.0240	-5.01***	0.00	-4.64***	0.00
ITA	0.86	-0.0842	0.0082	-10.29***	0	0.0198	-4.25***	0.00	-3.93***	0.00
JPN	0.82	-0.1573	0.0184	-8.54***	0	0.0427	-3.68***	0.00	-3.41***	0.00
NLD	0.87	-0.0923	0.0068	-13.67***	0	0.0207	-4.46***	0.00	-4.13***	0.00
USA	0.82	-0.0930	0.0094	-9.84***	0	0.0248	-3.75***	0.00	-3.47***	0.00
AGG	0.90	-0.0753	0.0042	-17.92***	0	0.0143	-5.26***	0.00	-4.87***	0.00

Notes: The t-statistics and implied p-values are calculated based on Newey–West's (1987) heteroskedasticity- and autocorrelation-consistent (HAC) standard errors. The number of lags is computed based on the non-parametric bandwidth selection procedure suggested by Newey and West (1994). Weights are computed using the Bartlett kernel as described in Newey and West (1987, 1994). Standard errors as well as t-statistics and p-values computed based on Harri and Brorsen (2009) are denoted by superscript $^{H+B}$, and t-statistics and p-values based on Hjalmarsson (2008, 2011) are denoted by superscript ^{H}. Statistical significance at the 1% level is indicated by ***, at the 5% level by ** and at the 10% level by *. *Abbreviations:* AUS = Australia, AUT = Austria, BEL = Belgium, CAN = Canada, CHE = Switzerland, DEU = Germany, DNK = Denmark, FRA = France, GBR = Great Britain, IRL = Ireland, ITA = Italy, JPN = Japan, NLD = Netherlands, USA = United States, AGG = equal-weighted country aggregate.

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Table 20: Regressing 15-yr forward nominal returns on ln(MVE/GDP), 1973-2019

	R^2	coef.	s.e.	t-stat.	p-value	s.e.$^{H+B}$	t-stat.$^{H+B}$	p-value^{H+B}	t-stat.^{H}	p-value^{H}
AUS	0.91	-0.0645	0.0028	-22.91***	0	0.0137	-4.71***	0.00	-4.51***	0.00
AUT	0.68	-0.0402	0.0042	-9.53***	0	0.0180	-2.23**	0.03	-2.14**	0.03
BEL	0.80	-0.0680	0.0069	-9.83***	0	0.0223	-3.05***	0.00	-2.92***	0.00
CAN	0.48	-0.0310	0.0066	-4.71***	0	0.0213	-1.45	0.15	-1.39	0.17
CHE	0.63	-0.0496	0.0086	-5.8***	0	0.0249	-1.99**	0.05	-1.91*	0.06
DEU	0.63	-0.0567	0.0082	-6.91***	0	0.0283	-2.00**	0.05	-1.91*	0.06
DNK	0.69	-0.0425	0.0048	-8.9***	0	0.0187	-2.27**	0.02	-2.18**	0.03
FRA	0.89	-0.0680	0.0046	-14.8***	0	0.0157	-4.32***	0.00	-4.13***	0.00
GBR	0.94	-0.1284	0.0069	-18.5***	0	0.0210	-6.11***	0.00	-5.84***	0.00
IRL	0.92	-0.1318	0.0072	-18.19***	0	0.0258	-5.11***	0.00	-4.89***	0.00
ITA	0.90	-0.0894	0.0062	-14.52***	0	0.0198	-4.51***	0.00	-4.31***	0.00
JPN	0.81	-0.1617	0.0181	-8.93***	0	0.0511	-3.16***	0.00	-3.03***	0.00
NLD	0.89	-0.1035	0.0066	-15.59***	0	0.0233	-4.43***	0.00	-4.24***	0.00
USA	0.83	-0.1053	0.0107	-9.82***	0	0.0318	-3.31***	0.00	-3.17***	0.00
AGG	0.90	-0.0836	0.0052	-16.21***	0	0.0185	-4.52***	0.00	-4.32***	0.00

Notes: The t-statistics and implied p-values are calculated based on Newey–West's (1987) heteroskedasticity- and autocorrelation-consistent (HAC) standard errors. The number of lags is computed based on the non-parametric bandwidth selection procedure suggested by Newey and West (1994). Weights are computed using the Bartlett kernel as described in Newey and West (1987, 1994). Standard errors as well as t-statistics and p-values computed based on Harri and Brorsen (2009) are denoted by superscript $^{H+B}$, and t-statistics and p-values based on Hjalmarsson (2008, 2011) are denoted by superscript ^{H}. Statistical significance at the 1% level is indicated by ***, at the 5% level by ** and at the 10% level by *. *Abbreviations:* AUS = Australia, AUT = Austria, BEL = Belgium, CAN = Canada, CHE = Switzerland, DEU = Germany, DNK = Denmark, FRA = France, GBR = Great Britain, IRL = Ireland, ITA = Italy, JPN = Japan, NLD = Netherlands, USA = United States, AGG = equal-weighted country aggregate.

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